

HOW TO BECOME A SCIENTIST



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Giving interesting and instructive experiments in chemistry, mechanics, acoustics and pyrotechnics

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How to Become a Scientist.

GIVING

Interesting and Instructive Experiments

IN

CHEMISTRY,
Mechanics, Acoustics

AND

PYROTECHNICS.

ALSO CONTAINING

MATHEMATICAL PROBLEMS and PUZZLES

BOTH

USEFUL AND AMUSING.

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How to Become a Scientist.

Chemistry, optics, pneumatics, mechanics, and mathematics, all contribute their share towards furnishing recreation and sport for the social gathering, or the family fireside. The magical combinations and effects of chemistry have furnished an almost infinite variety of pleasant experiments, which may be performed by our youthful friends with great success if a little care be taken; and the other branches of natural science are nearly as replete with interest.

The following *repertoire* of such tricks and illusions will be found exceedingly complete, although pains have been taken to select only the best and most startling of them. A large number are entirely new, but are described with sufficient clearness to enable any person of ordinary intelligence to become expert in them, with a little practice.

Chemical Amusements.

Chemistry is one of the most attractive sciences. From the beginning to the end the student is surprised and delighted with the developments of the exact discrimination, as well as the power and capacity, which are displayed in various forms of chemical action. Dissolve two substances in the same fluid, and then, by evaporation or otherwise, cause them to reassume a solid form, and each particle will unite with its own kind, to the entire exclusion of all others. Thus, if sulphate of copper and carbonate of soda are dissolved in boiling water, and then the water is evaporated, each salt will be reformed as before. This phenomenon is the result of one of the first principles of the science, and as such is passed over without thought; but it is a wonderful phenomenon, and made of no account, only by the fact that it is so common and so familiar.

It is by the action of this same principle, “chemical affinity,” that we produce the curious experiments with

Sympathetic Inks.

By means of these, we may carry on a correspondence which is beyond the discovery of all not in the secret. With one class of these inks, the writing becomes visible only when moistened with a particular solution. Thus, if we write to you with a solution of the sulphate of iron, the letters are invisible. On the receipt of our letter, you rub over the sheet a feather or sponge, wet with a solution of nut-galls, and the letters burst forth into sensible being at once, and are permanent.

2. If we write with a solution of sugar of lead, and you moisten with a sponge or pencil, dipped in water impregnated with sulphureted hydrogen, the letters will appear with metallic brilliancy.

3. If we write with a weak solution of sulphate of copper, and you apply ammonia, the letters assume a beautiful blue. When the ammonia evaporates, as it does on exposure to the sun, the writing disappears, but may be revived again as before.

4. If you write with the oil of vitriol very much diluted, so as to prevent its destroying the paper, the manuscript will be invisible except when held to the fire, when the letters will appear black.

5. Write with cobalt dissolved in diluted muriatic acid; the letters will be invisible when cold, but when warmed they will appear a bluish green.

We are almost sure that our secrets thus written will not be brought to the knowledge of a stranger, because he does not know the solution which was used in writing, and, therefore, does not know what to apply to bring out the letters.

To Light a Candle Without Touching the Wick.

Let the candle burn until it has a good long snuff; then blow it out with a sudden puff, a bright wreath of white smoke will curl up from the hot wick. Now, if a flame be applied to this smoke, even at a distance of two or three inches from the candle, the flame will run down the smoke and rekindle the wick in a very fantastic manner. To perform this experiment nicely, there must be no draught or “banging” doors while the mystic spell is rising.

Magic Milk.

Lime-water is quite transparent, and clear as common spring water; but if we breathe or blow into it, the bright liquid becomes opalescent and as white as milk.

The best way to try this simple experiment is to put some powdered quicklime into a wine bottle full of cold water; shake them well together, now and then, for a day; then allow the bottle to remain quiet till the next day, when the clear lime-water may be poured off from the sediment. Now fill a wine-glass or tumbler with the lime-water thus made, and blow through the liquid with a glass tube, a piece of new tobacco-pipe, or a clean straw, and in the course of a minute or so—as the magicians say—“the water will be turned into milk.” By means of this pastime “Wise Men” can ascertain which young ladies are in love and which young gentlemen are not. With a shrewd guess they present, as a test, a glass of lime-water to the one and of pure water to the other, with unerring effect.

The Mimic Vesuvius.

This experiment is a demonstration of the heat and light which are evolved during chemical combination. The substance phosphorus has a great affinity for oxygen gas, and wherever it can get it from it will, especially when aided by the application of heat. To perform this experiment, put half a drachm of solid phosphorus into a Florence oil-flask, holding the glass slantingly, that the phosphorus may not take fire and break the glass; pour upon it a gill and a half of water, and place the whole over a tea-kettle lamp, or any common lamp filled with spirits of wine; light the wick, which should be about half an inch from the flask; and as soon as the water is boiling hot, streams of fire, resembling sky-rockets, will burst at intervals from the water; some particles will also adhere to the sides of the glass, immediately displaying brilliant rays, and thus continue until the water begins to simmer, when a beautiful imitation of the aurora borealis will commence and gradually ascend until it collects into a pointed cone at the mouth of the flask; after a half a minute, blow out the flame of the lamp, and the apex of fire that was formed at the mouth of the flask will rush down, forming beautiful illumined clouds of fire, rolling over each other for some time; and when these disappear, a splendid hemisphere of stars will present itself. After waiting a minute or two, light the lamp again, and nearly the same phenomena will be displayed as at the beginning. Let a

repetition of lighting and blowing out the lamp be made for three or four times, so that the number of stars may be increased; and after the third or fourth act of blowing out the lamp, the internal surface of the flask will be dry. Many of the stars will shoot with great splendor from side to side, while others will appear and burst at the mouth of the flask. What liquid remains in the flask will serve for the same experiment three or four times, without adding any water. Care should be taken, after the operation is over, to put the flask in a cool and secure place.

The Real Will-o'-the-Wisp.

Into a small retort place about an ounce of strong liquor of potash; that is, pure potash dissolved in water, together with about a drachm of phosphorus. Let the neck or beak of the retort dip into a saucer of water, say half an inch deep; now very gently heat the liquid in the retort with a spirit-lamp until it boils. In a few minutes the retort will be filled with a white cloud; then the gas generated will begin to bubble at the end of the saucer; a minute more, each bubble, as it issues from the boiling fluid, will *spontaneously take fire* as it comes into the air, forming at the same time the philosopher's ring of phosphoric acid. Care is required in handling phosphorus; but our young chemical readers will, we think, not forego this wonderful experiment for the want of due attention; for, without proper care on their part, we must give up showing them wonders even greater than these.

The Paper Oracle.

Some amusement may be obtained among young people by writing, with common ink, a variety of questions, on different bits of paper, and adding a pertinent reply to each, written with nitro-muriate of gold. The collection should be suffered to dry, and put aside, until an opportunity offers for using them. When produced, the answers will be invisible; desire different persons to select such questions as they may fancy, and take them home with them; then promise, if they are placed near the fire during the night, answers will appear written beneath the questions in the morning; and such will be the fact, if the paper be put in any dry, warm situation.

The Mimic Gas-House.

This shows a simple way of making illuminating gas, by means of a tobacco-pipe. Bituminous coal contains a number of chemical compounds, nearly all of which can, by distillation, be converted into an illuminating gas; as with this gas nearly all our cities are now lighted in the dark hours of night. To make it, obtain some coal-dust (or walnut or butternut meats will answer), and fill the bowl of a pipe with it; then cement the top over with some clay; place the bowl in the fire, and soon smoke will be seen issuing from the end of the stem; when that has ceased coming apply a light and it will burn brilliantly for several minutes; after it has ceased, take the pipe from the fire and let it cool, then remove the clay, and a piece of coke will be found inside: this is the excess of carbon over the hydrogen contained in the coal, for all the hydrogen will combine with carbon at a high temperature, and make what are called hydrocarbons—a series of substances containing both these elemental forms of matter.

Alum Basket.

Make a small basket, about the size of the hand, of iron wire or split willow; then take some lamp-cotton, untwist it, and wind it around every portion of the basket. Then mix alum, in the proportion of one pound with a quart of water, and boil it until the alum is dissolved. Pour the solution into a deep pan, and in the liquor suspend the basket, so that no part of it touch the vessel or be exposed to the air. Let the whole remain perfectly at rest for twenty-four hours; when, if you take out the basket, the alum will be found prettily crystallized over all the limbs of the cottoned frame.

In like manner, a cinder, a piece of coke, the sprig of a plant, or any other object, suspended in the solution by a thread, will become covered with beautiful crystals.

If powdered tumeric be added to the hot solution, the crystals will be of a bright yellow; if litmus be used instead, they will be of a bright red; logwood will yield them of a purple, and common writing-ink, of a black tint; or, if sulphate of copper be used instead of alum, the crystals will be of fine blue.

But the colored alum crystals are much more brittle than those of pure alum, and the colors fly; the best way of preserving them is to place them under a glass shade, with a saucer containing water. This keeps the

atmosphere constantly saturated with moisture, the crystals never become too dry, and their texture and color undergo but little change.

The Magic Bottle.

This trick, if well managed, is one of the most wonderful that can be performed in a drawing-room without apparatus; but it requires dexterity at the conclusion.

The person performing the trick offers to pour from a common wine-bottle, port-wine, sherry, milk, and champagne, in succession, and in any order.

To accomplish the trick, you must make solutions of the following chemicals, and label the bottles with numbers, thus:

No. 1. A mixture of two parts perchloride of iron, and one part sulphuric acid (vitriol).

No. 2. A strong solution of the sulphocyanate of potash.

No. 3. A strong solution of acetate of lead.

No. 4. A solution of bicarbonate of soda, or potash.

No. 5. A clear solution of gum arabic.

Procure a champagne-bottle, and wash it out well; then pour three teaspoonfuls of No. 1 into it. As the quantity is very small, it will not be observed, especially if you are quick in your movements. Pour some distilled or rain water into a common water-bottle, or jug, and add a tablespoonful of No. 5 to it; then set it aside, ready for use.

Provide some wine-glasses, of four different patterns, and into one pattern put one drop of solution No. 2; into another, three drops of solution No. 2; rinse the third with solution No. 3, and the fourth with solution No. 4.

Arrange the glasses on a small tray, remembering the solutions that were poured into each pattern.

Everything being ready, take the champagne bottle that you have prepared, from two or three others, and holding it up, to show the company that it is clear and empty; you must desire some person to hand you the water-bottle or jug, and then fill up the bottle with the water.

Pour some of the contents of the bottle into an unprepared glass, in order to show that it is water; then say: “Change to champagne,” and pour the liquid from the bottle into one of the glasses rinsed with No. 4; then pour into the glass containing *three drops* of No. 2, and it will change to port wine; but if poured into the glass rinsed with No. [3](#), it will change to milk; and if into the glass with one drop of No. 2, it will produce sherry.

Be careful in pouring the fluid from the bottle, not to hold it high above the glasses, but to keep the mouth of it close to the edges, otherwise persons will observe that it undergoes change of color after it is poured into them; and, on this account, the glasses should be held rather high.

As all the solutions used in the above trick are deleterious, they should not be left about in the way of children, and, of course, the fluid in the wine-glasses must not even be tasted; but if any of the company wish to drink the wines you have made, then the tray must be adroitly exchanged for another with the proper wines placed on it.

The Faded Rose Restored.

Take a rose that is quite faded, and throw some sulphur on a chafing-dish of hot coals; then hold the rose over the fumes of the sulphur, and it will become quite white; in this state dip it into water, put it into a box, or drawer, for three or four hours, and when taken out it will be quite red again.

The Protean Liquid.

A red liquor, which, when poured into different glasses, will become yellow, blue, black, and violet, may be thus made: Infuse a few shavings of logwood in common water, and when the liquor is red, pour it into a bottle; then take three drinking-glasses, rinse one of them with strong vinegar, throw into the second a small quantity of pounded alum, which will not be observed if the glass has been newly washed, and leave the third without any preparation. If the red liquor in the bottle be poured into the first glass it will assume a straw color; if into the second, it will pass gradually from bluish-gray to black, provided it be stirred with a bit of iron, which has been privately immersed in good vinegar; in the third glass the red liquor will assume a violet tint.

The Changeable Ribbon.

Dip a rose-colored ribbon into nitric acid, diluted with eight or ten parts of water, and as soon as the color disappears, which it will do in a short time, take out the ribbon and put it into a very weak alkaline solution, when the alkali will quickly neutralize the acid, and the color will reappear.

The Chemical Chameleon.

Put a drachm of powdered nitrate of cobalt into a vial, containing an ounce of the solution of caustic potash, when the decomposition of the salt, and precipitation of a blue oxide of cobalt will take place. Cork the vial, and the liquid will assume a blue color, from which it will pass to a lilac, afterward to a peach tint, and finally to a light red.

Musical Flame.

Fit a good cork into a wine-bottle; burn a hole through the cork with a round iron skewer, and into it fix a piece of tobacco pipe about eight inches long. Put into the bottle about two or three ounces of zinc, in slips, such as the waste cuttings from a zinc-worker; now pour water on to the zinc until the bottle is more than half full; then add about three parts of a wine-glassful of sulphuric acid (oil of vitriol); this causes a rapid effervescence at first, but which subsides to a moderate and continuous boiling for a lengthened period; as soon as the boiling is regular, the cork with the pipe through it may be inserted into the bottle. If a light be placed to the end of the pipe, a flame will be produced, which will continue to burn so long as there is any visible action in the bottle. This flame is the ignited hydrogen gas (water gas), resulting from the decomposition of water by the acid and zinc, and as such is an exceedingly interesting experiment. Now, to be musical, procure a glass or metal pipe, about sixteen or eighteen inches long, and from half to three-quarters of an inch in diameter; place the tube over the flame, and allow the pipe to be about three to five inches up the tube, which will act as a kind of high chimney; it must be held perfectly steady and upright, at a particular distance up the tube, which varies according to the size of the flame. A beautiful sound is thus produced, similar to an organ-pipe. This sound, or “musical flame,” varies in note according to the diameter of the tube, being deeper or more bass as the tube

is increased in size. By using various-sized tubes, different sounds are thus readily produced. The true explanation of this singular experiment remains yet to be solved.

Optical Amusements.

The science of optics affords an infinite variety of amusements, which cannot fail to instruct the mind, as well as delight the eye. By the aid of optical instruments we are enabled to lessen the distance to our visual organs between the globe we inhabit and “the wonders of the heavens above us;” to watch “the stars in their courses,” and survey at leisure the magnificence of “comets importing change of times and states;” to observe the exquisite finish and propriety of construction which are to be found in the most minute productions of the earth;—to trace the path of the planet, in its course around the magnificent orb of day, and to detect the pulsation of the blood, as it flows through the veins of an insect. These are but a few of the powers which this science offers to man; to enumerate them all would require a space equal to the body of our work; neither do we propose to notice the various instruments and experiments which are devoted to purposes merely scientific; it being our desire only to call the attention of our juvenile readers to such things as combine a vast deal of amusement with much instruction, to inform them as to the construction of the various popular instruments; to show the manner of using them, and to explain some of the most attractive experiments which the science affords. By doing thus much, we hope to offer a sufficient inducement to extend inquiry much further than the information which a work of this nature will enable us to afford.

The Camera Obscura.

This is a very pleasing and instructive optical apparatus, and may be purchased for a small sum. But it may be easily made by the young optician. Procure an oblong box, about two feet long, twelve inches wide, and eight high. In one end of this a tube must be fitted containing a lens, and be made to slide backward and forward, so as to suit the focus. Within the box should be a plain mirror, reclining backward from the tube at an angle of forty-five degrees. At the top of the box is a square of unpolished glass, upon which, from beneath, the picture will be thrown, and may be

seen by raising the lid. To use the camera, place the tube with the lens on it opposite to the object, and having adjusted the focus, the image will be thrown upon the ground glass, as above stated, where it may be easily copied by a pencil or in colors.

The Magic Lantern.

The object of this ingenious instrument is to represent, in a dark room, on a white wall or cloth, a succession of enlarged figures of remarkable, natural, or grotesque objects. It consists of a tin box, with a funnel on the top, and a door on one side of it. This funnel, by being bent, serves the double purpose of letting out the smoke and keeping in the light. In the middle of the bottom of the box is placed a movable lamp, which must have two or three good lights, at the height of the center of the polished tin reflector. In the front of the box, opposite the reflector, is fixed a tin tube, in which there slides another tube. The sliding tube has, at its outer extremity, a convex lens fixed in it, of three inches in diameter. The focus of the smaller of these lenses may be about five inches. Between the stationary tube and the lamp, there must be a split or opening to admit of the passage of glass sliders, mounted in paper or wooden frames, upon which sliders it is that the miniature figures are painted, which are intended to be shown upon the wall. The distinctness of the enlarged figures depends not only upon the goodness of the magnifying glass, but upon the clearness of the light yielded by the lamp. It may be purchased ready made of any optician.

To Paint the Glasses.—The slides containing the objects usually shown in a magic lantern are to be bought of opticians with the lantern, and can be procured cheaper and better in this way than by any attempt at manufacturing them. Should, however, the young optician wish to make a few slides, of objects of particular interest to himself, he may proceed as follows: Draw on a paper the subject you desire to paint. Lay it on a table or any flat surface, and place the glass over it; then draw the outlines with a very fine pencil, in varnish mixed with black paint, and, when dry, fill up the other parts in their proper colors. Transparent colors must be used for this purpose, such as carmine, lake, Prussian blue, verdigris, sulphate of iron, tincture of Brazil wood, gamboge, etc.; and these must be tempered with a strong white varnish, to prevent their peeling off. Then shade them with black, or with bistre, mixed with the same varnish.

To Exhibit the Magic Lantern.—The room for the exhibition ought to be large, and of an oblong shape. At one end of it suspend a large sheet, so as to cover the whole of the wall. The company being all seated, darken the room, and placing the lantern with its tube in the direction of the sheet, introduce one of the slides into the slit, taking care to invert the figures; then adjust the focus of the glasses in the tube, by drawing it in or out, as required, and a perfect representation of the object will appear.

Effects of the Magic Lantern.—Most extraordinary effects may be produced by means of the magic lantern; one of the most effective of which is a tempest at sea.

This is effected by having two slides painted, one with the tempest as approaching on one side, and continuing in intensity till it reaches the other. Another slide has ships painted on it, and while the lantern is in use, that containing the ships is dexterously drawn before the other, and represents ships in the storm.

The effects of sunrise, moonlight, starlight, etc., may be imitated also, by means of double sliders; and figures may be introduced sometimes of fearful proportions.

Heads may be made to nod, faces to laugh; eyes may be made to roll, teeth to gnash; crocodiles may be made to swallow tigers; combats may be represented; but one of the most instructive uses of the slides is to make them illustrative of astronomy, and to show the ratio of the seasons, the cause of the eclipses, the mountains in the moon, spots on the sun, and the various motions of the planetary bodies and their satellites.

The Phantasmagoria.

Between the phantasmagoria and the magic lantern there is this difference: in common magic lanterns the figures are painted on transparent glass; consequently the image on the screen is a circle of light, having figures upon it; but in the phantasmagoria all the glass is opaque, except the figures, which, being painted in transparent colors, the light shines through them, and no light can come upon the screen except that which passes through the figure.

There is no sheet to receive the picture, but the representation is thrown on a thin screen of silk or muslin, placed between *the spectators and the lantern*. The images are made to appear approaching and receding, by removing the lantern further from the screen, or bringing it nearer to it. This is a great advantage over the arrangements of the magic lantern, and by it the most astonishing effects are often produced.

Dissolving Views.

The dissolving views, by which one landscape or scene appears to pass into the other while the scene is changing, are produced by using two magic lanterns, placed side by side, and that can be inclined towards each other when necessary, so as to mix the rays of light, proceeding from the lenses of each, together, which produces that confusion of images, in which one view melts, as it were, into the other, which gradually becomes clear and distinct.

How to Raise a Ghost.

The magic lantern or phantasmagoria may be used in a number of marvelous ways, but in none more striking than in raising an apparent specter. Let an open box, about three feet long, a foot and a half broad, and two feet high, be prepared. At one end of this place a small swing dressing-glass, and at the other let a magic lantern be fixed, with the lenses in a direction towards the glass. A glass should now be made to slide up and down in the groove to which a cord and pulley should be attached, the end of the cord coming to the lower part of the left hand side. On this glass the most hideous specter that can be imagined may be painted, but in a squat or contracted position, and when all is done, the lid of the box must be prepared by raising a kind of gable at the end of the box, and in its lower part an oval hole should be cut sufficiently large to suffer the rays reflected from the glass to pass through them. On the top of the box place a chafing-dish, upon which put some burning charcoal. Now light the lamp in the lantern, sprinkle some powdered camphor or white incense on the charcoal, adjust the slide on which the specter is painted, and the image will be thrown upon the smoke. In performing the feat the room must be darkened, and the box should be placed on a high table, that the hole through which the light comes may not be noticed.

To Imitate a Mirage.

Provide a glass tumbler two-thirds full of water, and pour spirits of wine upon it; or pour into a tumbler some syrup, and fill it up with water; when mixed, the object seen through it will be inverted.

Two-fold Reflections.

Provide a circular piece of glass, and with a common awl, moistened with spirits of turpentine, pierce the center of the glass; hold it encircled with the fingers and thumb in the sunshine, or the strong light of a lamp, when these striking effects will be produced: If the glass be *red*, the hole pierced in the middle will be reflected *green*; if the glass be *green*, the spot will be *red*; if *blue*, *orange*; and if *yellow*, *indigo*.

The Thaumatrope.

Cut out a piece of card-board of circular form, and affix to it six pieces of string, three on each side. Paint on one side of the card a bird, and on the other a cage, taking care to paint the bird upside down, or the desired effect will not be produced. When showing the toy, take hold of the center strings, between the forefinger and thumb, and twirl the card rapidly around, and the bird will appear snugly ensconced in its cage. The principle on which this effect is produced is, that the image of any object received on the retina or optic nerve is retained on the mind about eight seconds after the object causing the impression is withdrawn, being the memory of the object; consequently, the impression of the painting on one side of the card is not obliterated ere the painting on the other side is brought before the eye. It is easy to understand from this fact how both are seen at once. Many objects will suit the thaumatrope, such as a juggler throwing up two balls on one side, and two balls on the other; and according to the pairs of strings employed, he will appear to throw up two, three, or four balls; the body and legs of a man on one side, and the arms and head on another; a horse and his rider; a mouse and trap. But we leave it to the ingenuity of our readers to devise for themselves.

PNEUMATIC AMUSEMENTS.

The branch of the physical sciences which relates to the air and its various phenomena is called Pneumatics. By it we learn many curious particulars. By it we find that the air has weight and pressure, color, density, elasticity, compressibility, and some other properties with which we shall endeavor to make the young reader acquainted, by many pleasing experiments, earnestly impressing upon him to lose no opportunity of making physical science his study.

To show that the air has weight and pressure, the common leather sucker by which boys raise stones will show the pressure of the atmosphere. It consists of a piece of soft but firm leather, having a piece of string drawn through its center. The leather is made quite wet and pliable, and then its under part is placed upon the stone and stamped down by the foot. This pressing of the leather excludes the air from between the leather and the stone, and by pulling the string a vacuum is left underneath its center; consequently the weight of the air about the edges of the leather not being counterbalanced by any air between it and the stone, enables the boy to lift it.

The Magic Tumbler.

The air which for about forty miles surrounds our earth has a definite weight; and although we can neither see nor feel it, we are conscious of its presence by the momentary operation of breathing. The weight of a column of air one inch square, and forty miles high, is about fifteen pounds.

The reason why we are not crushed down by this enormous weight is because we are surrounded on all sides by it, and as the pressure of weight is equal all around, it becomes, as far as we are personally concerned, insensible.

That the air *does* exert a definite pressure, in consequence of its weight, may be easily proved by any one with the above simple apparatus—only a tumbler and a sheet of paper. Fill a tumbler quite full of water, and carefully draw over its top a sheet of clean letter paper, and be careful to see that there are no bubbles of air in the water; place your hand over the paper

while inverting it, and when the glass is mouth downward the water will be kept in, until the paper becomes wet through. The air pressing against the mouth of the tumbler is of greater weight than the contained water, and so, until some air can get in to supply the place of the water, it cannot fall out.

The Weight of the Air Proved by a Pair of Bellows.

Shut the nozzle and valve-hole of a pair of bellows, and after having squeezed the air out of them, if they are perfectly air-tight, we shall find that a very great force, even some hundreds of pounds, is necessary for separating the boards. They are kept together by the weight of the heavy air which surrounds them, in the same manner as if they were surrounded by water.

The Revolving Serpent.

[This illustration](#) represents an amusing and instructive experiment, which proves the ascension of heated air by rendering its effects visible, and it may also be used to test the direction of the currents in our rooms and dwellings. To construct one, a piece of card-board is taken and cut in the form of a spiral, and to give effect it may be painted to represent a serpent. Then prepare a stand, having a needle in its upper end, and suspend the serpent from its center on the needle. If this be now placed over a stove, or the tail of the serpent suspended by a bit of thread over a lamp, the heated air ascending through it will cause it to revolve in a very amusing manner. Two serpents may be made to turn in opposite directions, by pulling out one from the one side, and the other in the reverse direction, so that their heads may point toward each other when suspended.

To Put a Lighted Candle Under Water.

Procure a good-sized cork, or bung; upon this place a small, lighted taper; then set it afloat in a pail of water. Now, with a steady hand, invert a large drinking glass over the light, and push it carefully down into the water. The glass being full of air, prevents the water from entering it. You may thus see the candle burn *under* water, and bring it up again to the surface, still alight. This experiment, simple as it is, serves to elucidate that useful contrivance called the diving-bell, being performed on the same principle.

The largest drinking-glass holds but half a pint, so that your diving light soon goes out for the want of air. As an average, a burning candle consumes as much air as a man, and he requires nearly a gallon of air every minute, so that, according to the size of the glass over the flame, you can calculate how many seconds it will remain alight; of course, a large flame requires more air than a small one. For this, and several other experiments, a quart bell-glass is very useful, but being expensive it is not found in every parlor laboratory: one is, however, easily made from a green glass pickle-bottle; get a glazier to cut off the bottom, and you have a bell-glass that Chilton would not reject.

To Place Water in a Drinking-Glass Upside Down.

Procure a plate, a tumbler, and a small piece of tissue or silver paper. Set the plate on a table, and pour water in it up to the first rim. Now slightly crumple up the paper, and place it in the glass; then set it on fire. When it is burnt out, or rather just as the last flame disappears, turn the glass quickly upside down into the water. Astonishing! the water rushes with great violence into the glass! Now you are satisfied that water can be placed in a drinking-glass upside down. Hold the glass firm, and the plate also. You can now reverse the position of the plate and glass, and thus convince the most skeptical of the truth of your pneumatic experiment. Instead of burning paper, a little brandy or spirits of wine can be ignited in the glass; the result of its combustion being invisible, the experiment is cleaner.

AMUSEMENTS IN MECHANICS.

There is no subject so important as mechanics, as its principles are founded upon the properties of matter and the laws of motion; and, knowing something of these, the tyro will lay the foundation of all substantial knowledge.

The properties of matter are the following: Solidity (or impenetrability), divisibility, mobility, elasticity, brittleness, malleability, ductility and tenacity.

The laws of motion are as follows:

1. Every body continues in a state of rest, or uniform rectilinear motion, unless affected by some extraneous force.
2. The change of motion is always proportionate to the impelling force.
3. Action and reaction are always equal and contrary.

Experiment of the Law of Motion.

In shooting at “taw,” if the marble be struck “plump,” as it is called, it moves forward exactly in the same line of direction; but if struck sideways, it will move in an oblique direction, and its course will be in a line situated between the direction of its former motion and that of the force impressed. It is called the resolution of forces.

Balancing.

The center of gravity in a body is that part about which all the other parts equally balance each other. In balancing a stick upon the finger, or upon the chin, it is necessary only to keep the chin or finger exactly under the point which is called the center of gravity.

The Balanced Coin.

It seems to be an astounding statement that a quarter, or other piece of money, can be made to spin on the point of a needle. To perform this experiment, procure a bottle, cork it, and in the cork place a needle. Now

take another cork and cut a slit in it, so that the edge of the coin will fit into the slit; next place two forks in the cork, and placing the edge of the coin on the needle, it will spin around without falling off. The reason is this: that the weight of the forks projecting as they do so much below the coin, brings the center of gravity of the arrangement much below the point of suspension, or the point of the needle, and therefore the coin remains perfectly safe and upright.

The Spanish Dancer.

The laws which govern the motion of bodies are capable of many pleasing illustrations, and the example which we now give of causing rotary motion is very interesting and easily performed.

Take a piece of card, and cut out a little figure, and paste or gum it in an erect position on the inside of a watch-glass. Then procure a black japanned waiter, or a clean plate will do, and, holding it in an inclined position, place the figure and watch-glass on it, and they will, of course, slide down. Next let fall a drop of water on the waiter, place the watch-glass on it, and again incline the waiter, and instead of the watch-glass sliding down, it will begin to revolve. It will continue to revolve with increasing velocity, obeying the inclination and position of the plane, as directed by the hand of the experimentalist. The reason of this is, in the first place, in consequence of the cohesion of the water to the two surfaces, a new force is introduced, by which an unequal degree of resistance is imparted to different parts of the watch-glass in contact with the waiter, and, consequently, in its effort to slide down, it revolves. Again, if the drop of water be observed, it will be seen that it undergoes a change of figure; a film of water, by capillary action, is drawn to the foremost portion of the glass, while, by the centrifugal force, a body of water is thrown under the under part of it. The effect of both these actions is to accelerate the motion, or, in other words, to gradually increase the speed.

The Mechanical Bucephalus.

The illustration of the horse furnishes a very good solution of a popular paradox in mechanics: Given, a body having a tendency to fall by its own weight; required, how to prevent it from falling by adding to it a weight on

the same side on which it tends to fall. Take a horse in an erect position, the center of gravity of which is somewhere about the middle of its body. It is evident, therefore, that were it placed on its hinder legs, on a table, the line of its direction, or center, would fall considerably beyond its base, and the horse would fall on the ground; but to prevent this, there is a stiff wire attached to a weight or bullet, connected with the body of the horse, and by this means a horse prances on a table without falling off; so that the figure that was incapable of supporting itself, is actually prevented from falling by adding a weight to its unsupported end. This seems almost impossible, but when we consider that in order to have the desired effect, the wire must be bent, and the weight be further under the table than the horse's feet are on it, the mystery is solved, as it brings the total weight of bullet and horse in such a position that the tendency is rather to make it stand up than to let it fall down.

The Revolving Image.

This little figure may be made to balance itself amusingly. Get a piece of wood, about two inches long; cut one end of it into the form of a man's head and shoulders, and let the other end taper off to a fine point. Next furnish the little gentleman with a pair of wafers, shaped like oars, instead of arms, but they must be more than double the length of his body; stick them in his shoulders, and he is complete. When you place him on the tip of your finger, if you have taken care to make the point exactly in the center, he will stand upright. By blowing on the waiters he may be made to turn around very quickly. It is explained by the reasons that were given in the experiment of the "balanced coin."

The Bridge of Knives.

Place three glasses in the form of a triangle, and arrange the handles of three knives upon them. Nos. 1, 2, and 3, the blade of No. 1 over that of No. 2, and that over No. 3, which rests on No. 1. The bridge so made will be self-supported.

The Parlor Boomerang.

The boomerang is a weapon used by the savages of Australia. By them it is made of a flat piece of hard wood. The peculiarity of this instrument is, that in whatever direction it is thrown, it will return to the place from whence it started, in a curve. The Australian aborigines use it with great dexterity, making it travel around a house and return to their feet, or they can throw it on the ground so that it will fly into the air, form a perfect arc over their heads, and strike them on the back. This curious instrument can be made in miniature, and is a very amusing toy for the parlor.

Get a piece of tolerably stiff cardboard, and cut from it a figure resembling a boomerang.

The next thing is to propel it through the air so that it will return to your feet; to do this, lay the boomerang on a flat book, allowing one end to project about an inch; then, holding the book to a slight angle, strike the projecting end of the boomerang with a piece of stick, or heavy pen-holder, when it will fly across the room and return to your feet.

The Balanced Turk.

A decanter or bottle is first obtained, and in its cork is placed a needle; on this is balanced a ball of wood, having a cork or wooden figure cut out, standing on the top. From the ball project two wires, bent semicircularly, having at their extremities two bullets. Push the bullets, and the whole will turn around on the needle, the figure standing upright all the while; and, twist it about from side to side as much as you like, it will always regain its erect position. The two bullets in this case cause the center of gravity to fall below the ball on which the figure is placed, and, in consequence, as the center of gravity always assumes the lowest position, it cannot do so without making the figure stand erect, or, in other words, until the bullets themselves are equally balanced. Any boy may whittle one of these toys out with a jack-knife.

The Complacent Vizier.

Among the novelties which scientific investigation has added to our toys, are several figures which will raise themselves upright when thrown down, and regain the erect position, notwithstanding their equilibrium is disturbed. The figures themselves are made of the pith of elder trees, or any other very

light substance. Each is placed on half a bullet, or may be made to stand on its head, by making its cap of lead. Their appearance is very droll when they are moved about, as they seem every moment to be falling over, and yet continually right themselves. The philosophy of this is, that the center of gravity being in the base, and always trying to assume the lowest position, it keeps the figures upright. However much the equilibrium is disturbed, it will always try to regain its original position.

ARITHMETICAL AMUSEMENTS.

As the principal object of these articles is to enable the young reader to learn something in his sports, and to understand what he is doing, we shall, before proceeding to the curious tricks and feats connected with the science of numbers, present him with some arithmetical aphorisms, upon which most of the following examples are founded:

Aphorisms of Number.

1. If two even numbers be added together, or subtracted from each other, their sum or difference will be an even number.
2. If two uneven numbers be added or subtracted, their sum or difference will be an even number.
3. The sum or difference of an even and an uneven number added or subtracted will be an uneven number.
4. The product of two even numbers will be an even number, and the product of two uneven numbers will be an uneven number.
5. The product of an even and uneven number will be an even number.
6. If two different numbers be divisible by any one number, their sum and their difference will also be divisible by that number.
7. If several different numbers, divisible by 3, be added or multiplied together, their sum and their product will also be divisible by 3.
8. If two numbers divisible by 9, be added together, their sum of the figures in the amount will be either 9 or a number divisible by 9.
9. If any number be multiplied by 9, or by any other number divisible by 9, the amount of the figures of the product will be either 9 or a number divisible by 9.
10. In every arithmetical progression, if the first and last term be each multiplied by the number of terms, and the sum of the two products be divided by 2, the quotient will be the sum of the series.

11. In every geometric progression, if any two terms be multiplied together, their product will be equal to that term which answers to the sum of these two indices. Thus, in the series:

1	2	3	4	5
2	4	8	16	32

If the third and fourth terms, 8 and 16, be multiplied together, the product, 128, will be the seventh term of the series. In like manner, if the fifth term be multiplied into itself, the product will be the tenth term; and if that sum be multiplied into itself, the product will be the twentieth term. Therefore, to find the last, or twentieth term of a geometric series, it is not necessary to continue the series beyond a few of the first terms.

Previous to the numerical recreations, we shall here describe certain mechanical methods of performing arithmetical calculations, such as are not only in themselves entertaining, but will be found more or less useful to the young reader.

To Find a Number Thought of.

FIRST METHOD.

EXAMPLE.

Let a person think of a number, say	6
1. Let him multiply by 3	18
2. Add 1	19
3. Multiply by 3	57
4. Add to this the number thought of	63

Let him inform you what is the number produced; it will always end with 3. Strike off the 3, and inform him that he thought of 6.

SECOND METHOD.

EXAMPLE.

Suppose the number thought of to be	6
1. Let him double it	12
2. Add 4	16
3. Multiply by 5	80

- | | |
|-------------------|-----|
| 4. Add 12 | 92 |
| 5. Multiply by 10 | 920 |

Let him inform you what is the number produced. You must then, in every case, subtract 320; the remainder is, in this example, 600; strike off the 2 ciphers, and announce 6 as the number thought of.

THIRD METHOD.

Desire a person to think of a number—say 6. He must then proceed:

EXAMPLE.

- | | |
|--|----|
| 1. To multiply this number by itself | 36 |
| 2. To take 1 from the number thought of | 5 |
| 3. To multiply this by itself | 25 |
| 4. To tell you the difference between this product
and the former | 11 |
| You must then add 1 to it | 12 |
| And halve this number | 6 |

Which will be the number thought of.

FOURTH METHOD.

Desire a person to think of a number—say 6. He must then proceed as follows:

EXAMPLE.

- | | |
|---------------------------------------|----|
| 1. Add 1 to it | 7 |
| 2. Multiply by 3 | 21 |
| 3. Add 1 again | 22 |
| 4. Add the number thought of | 28 |
| Let him tell you the figures produced | 28 |
| 5. You then subtract 4 from it | 24 |
| 6. And divide by 4 | 6 |

Which you can say is the number he thought of.

FIFTH METHOD.

EXAMPLE.

- | | |
|----------------------------------|---|
| Suppose the number thought of be | 6 |
|----------------------------------|---|

- | | |
|---|----|
| 1. Let him double it | 12 |
| 2. Desire him to add to this a number you tell
him—say 4 | 16 |
| 3. To halve it | 8 |

You can then tell him that if he will subtract from this the number he thought of, the remainder will be, in the case supposed, 2.

NOTE.—The remainder is always half the number you tell him to add.

To Discover Two or More Numbers that a Person has Thought of.

FIRST CASE.

Where each of the numbers is less than 10. Suppose the numbers thought of were 2, 3, 5.

EXAMPLE.

- | | |
|--|----|
| 1. Desire him to double the first number, making | 4 |
| 2. To add one to it | 5 |
| 3. To multiply by 5 | 25 |
| 4. To add the second number | 28 |

There being a third number, repeat the process.

- | | |
|----------------------------|-----|
| 5. To double it | 56 |
| 6. To add 1 to it | 57 |
| 7. To multiply by 5 | 285 |
| 8. To add the third number | 290 |

And to proceed in the same manner for as many numbers as were thought of. Let him tell you the last sum produced (in this case, 290). Then, if there were two numbers thought of, you must subtract 5; if three, 55; if four, 555. You must here subtract 55; leaving a remainder of 235, which are the numbers thought of, 2, 3, and 5.

SECOND CASE.

Where one or more of the numbers are 10, or more than 10, and where there is an *odd* number of numbers thought of.

Suppose he fixes upon five numbers, viz., 4, 6, 9, 15, 16.

He must add together the numbers as follows, and tell you the various sums:

- | | |
|--------------------------------|----|
| 1. The sum of the 1st and 2d | 10 |
| 2. The sum of the 2d and 3d | 15 |
| 3. The sum of the 3d and 4th | 24 |
| 4. The sum of the 4th and 5th | 31 |
| 5. The sum of the 1st and last | 20 |

You must then add together the 1st, 3d, and 5th sums, viz., $10 + 24 + 20 = 54$, and the 2d and 4th, $15 + 31 = 46$; take one from the other, leaving 8. The half of this is the first number, 4; if you take this from the sum of the 1st and 2d you will have the 2d number, 6; this taken from the sum of the 2d and 3d will give you the 3d, 9; and so on for the other numbers.

THIRD CASE.

Where one or more of the numbers are 10, or more than 10, and where an even number of numbers has been thought of.

Suppose he fixes on six numbers, viz: 2, 6, 7, 15, 16, 18. He must add together the numbers as follows, and tell you the sum in each case:

- | | |
|-------------------------------|----|
| 1. The sum of the 1st and 2d | 8 |
| 2. The sum of the 2d and 3d | 13 |
| 3. The sum of the 3d and 4th | 22 |
| 4. The sum of the 4th and 5th | 31 |
| 5. The sum of the 5th and 6th | 34 |
| 6. The sum of the 2d and last | 24 |

You must then add together the 2d, 4th, and 6th sums, $13 + 31 + 24 = 68$, and the 3d and 5th sums, $22 + 34 = 56$. Subtract one from the other, leaving 12; the 2d number will be 6, the half of this; take the 2d from the sum of the 1st and 2d, and you will get the 1st; take the 2d from the sum of the 2d and 3d, and you will have the 3d, and so on.

How Many Counters Have I in My Hands?

A person having an equal number of counters in each hand, it is required to find how many he has altogether.

Suppose he has 16 counters, or 8 in each hand. Desire him to transfer from one hand to the other a certain number of them, and to tell you the number so transferred. Suppose it be 4, the hands now contain 4 and 12. Ask him how many times the smaller number is contained in the larger; in this case it is three times. You must then multiply the number transferred, 4, by the 3, making 12, and add the 4, making 16; then divide 16 by the 3 minus 1; this will bring 8, the number in each hand.

In most cases fractions will occur in the process; when 10 counters are in each hand and if four be transferred, the hands will contain 6 and 14.

He will divide 14 by 6 and inform you that the quotient is $2\frac{1}{3}$.

You multiply 4 by $2\frac{1}{3}$, which is $9\frac{1}{3}$.

Add four to this, making $13\frac{1}{3}$ equal to $\frac{40}{3}$.

Subtract 1 from $2\frac{1}{3}$, leaving $1\frac{1}{3}$ or $\frac{4}{3}$.

Divide $\frac{40}{3}$ by $\frac{4}{3}$, giving 10, the number in each hand.

The Three Travelers.

Three men met at a caravansary or inn, in Persia; and two of them brought their provisions along with them, according to the custom of the country; but the third, not having provided any, proposed to the others that they should eat together, and he would pay the value of his proportion. This being agreed to, A produced 5 loaves, and B 3 loaves, all of which the travelers ate together, and C paid 8 pieces of money as the value of his share, with which the others were satisfied, but quarreled about the division of it. Upon this the matter was referred to the judge, who decided impartially. What was his decision?

At first sight it would seem that the money should be divided according to the bread furnished; but we must consider that as the 3 ate 8 loaves, each one ate $2\frac{2}{3}$ loaves of the bread he furnished. This from 5 would leave $2\frac{1}{3}$ loaves furnished the stranger by A; and $3 - 2\frac{2}{3} = \frac{1}{3}$ furnished by B, hence $2\frac{1}{3}$ to $\frac{1}{3} = 7$ to 1, is the ratio in which the money is to be divided. If you

imagine A and B to furnish, and C to consume all, then the division will be according to amounts furnished.

The Money Game.

A person having in one hand a piece of gold, and in the other a piece of silver, you may tell in which hand he has the gold, and in which the silver, by the following method: Some value, represented by an even number, such as 8, must be assigned to the gold; and a value represented by an odd number, such as 3, must be assigned to the silver; after which, desire the person to multiply the number in the right hand by any even number whatever, such as 2, and that in the left by an odd number, as 3; then bid him add together the two products, and if the whole sum be odd, the gold will be in the right hand, and the silver in the left; if the sum be even, the contrary will be the case.

To conceal the artifice better, it will be sufficient to ask whether the sum of the two products can be halved without a remainder, for in that case the total will be even, and in the contrary case odd.

It may be readily seen that the pieces, instead of being in the two hands of the same person, may be supposed to be in the hands of two persons, one of whom has the even number, or piece of gold, and the other the odd number, or piece of silver. The same operations may then be performed in regard to these two persons, as are performed in regard to the two hands of the same person, calling the one privately the right, and the other the left.

The Philosopher's Pupils.

To find a number of which the half, fourth, and seventh, added to three, shall be equal to itself.

This was a favorite problem among the ancient Grecian arithmeticians, who stated the question in the following manner: "Tell us, illustrious Pythagoras, how many pupils frequent thy school?" "One-half," replied the philosopher, "study mathematics, one-fourth natural philosophy, one-seventh preserve silence, and there are three females besides."

The answer is 28: $14 + 7 + 4 + 3 = 28$.

The Certain Game.

Two persons agree to take, alternately, numbers less than a given figure, for example, 11, and to add them together till one of them has reached a certain sum, such as 100. By what means can one of them infallibly attain to that number before the other?

The whole artifice in this consists in immediately making choice of the numbers, 1, 12, 23, 34, and so on, or of a series which continually increases by 11, up to 100. Let us suppose that the first person, who knows the game, makes choice of 1, it is evident that his adversary, as he must count less than 11, can at most reach 11, by adding 10 to it. The first will then take 1, which will make 12; and whatever number the second may add, the first will certainly win, provided he continually add the number which forms the complement of that of his adversary to 11; that is to say, if the latter take 8, he must take 3; if 9, he must take 2; and so on. By following this method, he will infallibly attain to 89, and it will then be impossible for the second to prevent him from getting first to 100; for whatever number the second takes he can attain only to 99; after which the first may say—"and 1 makes 100." If the second take 1 after 89, it would make 90, and his adversary would finish by saying—"and 10 make 100." Between two persons who are equally acquainted with the game, he who begins must necessarily win.

The Dice Guessed Unseen.

A pair of dice being thrown, to find the number of points on each die without seeing them. Tell the person who cast the dice to double the number of points upon one of them, and add 5 to it; then to multiply the sum produced by 5, and to add to the product the number of points upon the other die. This being done, desire him to tell you the amount, and having thrown out 25, the remainder will be a number consisting of two figures, the first of which, to the left, is the number of points on the first die, and the second figure, to the right, the number of the other. Thus:

Suppose the number of points of the first die which comes up to be 2, and that of the other 3; then, if to 4, the double of the points of the first, there be added 5, and the sum produced, 9, be multiplied by 5, the product will be 45; to which, if 3, the number of points on the other die, be added, 48 will be produced, from which, if 25 be subtracted, 23 will remain; the first

figure of which is 2, the number of points on the first die, and the second figure 3, the number on the other.

The Famous Forty-five.

How can number 45 be divided into four such parts that, if to the first part you add 2, from the second part you subtract 2, the third part you multiply by 2, and the fourth part you divide by 2, the sum of the addition, the remainder of the subtraction, the product of the multiplication, and the quotient of the division, be all equal?

The first is	8;	to which add	2,	the sum is	10
The second is	12;	subtract	2,	the remainder is	10
The third is	5;	multiplied by	2,	the product is	10
The fourth is	<u>20;</u>	divided by	2,	the quotient is	10
	45				

Required to subtract 45 from 45, and leave 45 as a remainder.

SOLUTION.— $9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 = 45.$
 $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 = 45.$
 $8 + 6 + 4 + 1 + 9 + 7 + 5 + 3 + 2 = 45.$

The Astonished Farmer.

A and B took each 30 pigs to market. A sold his at 3 for a dollar, B at 2 for a dollar, and together they received \$25. A afterwards took 60 alone, which he sold as before, at 5 for \$2, and received but \$24: what became of the other dollar?

This is rather a catch question, the insinuation that the first lot were sold at the rate of 5 for \$2, being only true in part. They commence selling at that rate, but after making ten sales, A's pigs are exhausted, and they have received \$20; B still has 10, which he sells at "two for a dollar," and of course receives \$5; whereas had he sold them at the rate of 5 for \$2, he would have received but \$4. Hence the difficulty is easily settled.

The Expunged Figure.

In the first place we desire a person to write down secretly, in a line, any number of figures he may choose, and add them together as units; having done this, tell him to subtract that sum from the line of figures originally set down; then desire him to strike out any figure he pleases, and add the remaining figures in the line together as units (as in the first instance), and inform you of the result, when you will tell him the figure he has struck out.

$$\begin{array}{r} 76542-24 \\ \underline{24} \\ 76518 \end{array}$$

Suppose, for example, the figures put down are 76542; these added together, as units, make a total of 24; deduct twenty-four from the first line, and 76518 remain; if 5, the center figure, be struck out, the total will be 22. If 8, the first figure, be struck out, 19 will be the total.

In order to ascertain which figure has been struck out, you make a mental sum one multiple of 9 higher than the total given. If 22 be given as the total, then 3 times 9 are 27, and 22 from 27 show that 5 was struck. If 19 be given, that sum deducted from 27 shows 8.

Should the total be equal multiples of 9, as 18, 27, 36, then 9 has been expunged.

With very little practice, any person may perform this with rapidity: it is therefore needless to give any further examples. The only way in which a person can fail in solving this riddle is when either a number 9 or a 0 is struck out, as it then becomes impossible to tell which of the two it is, the sum of the figures in the line being an even number of nines in both cases.

Mysterious Addition.

It is required to name the quotient of five or three lines of figures—each line consisting of five or more figures—only seeing the first line before the other lines are even put down. Any person may write down the first line of figures for you. How do you find the quotient?

$$\begin{array}{r} 86,214 \\ 42,680 \\ 57,319 \end{array}$$

$$\begin{array}{r}
 62,854 \\
 37,145 \\
 \hline
 286,212
 \end{array}$$

When the first line of figures is set down, subtract 2 from the last right-hand figure, and place it before the first figure of the line, and that is the quotient for five lines. For example, suppose the figures are 86,214, the quotient will be 286,212. You may allow any person to put down the two first and the fourth lines, but you must always set down the third and fifth lines, and in doing so always make up 9 with the line above.

Therefore in the annexed diagram you will see that you have made 9 in the third and fifth lines with the lines above them. If the person you request to put down the figures should set down a 1 or 0 for the last figure, you must say: "We will have another figure," and another, and so on until he sets down something above 1 or 2.

$$\begin{array}{r}
 67,856 \\
 47,218 \\
 52,781 \\
 \hline
 167,855
 \end{array}$$

In solving the puzzle with 3 lines, you subtract 1 from the last figure, and place it before the first figure, and make up the third line yourself to 9. For example: 67,856 is given, and the quotient will be 167,855, as shown in the above diagram.

The Remainder.

A very pleasing way to arrive at an arithmetical sum, without the use of either slate or pencil, is to ask a person to think of a figure, then to double it, then add a certain figure to it, now halve the whole sum, and finally to abstract from that the figure first thought of. You are then to tell the thinker what is the remainder.

The key to this lock of figures is, that *half* of whatever sum you request to be added during the working of the sum *is the remainder*. In the example given, 5 is the half of 10, the number requested to be added. Any amount may be added, but the operation is simplified by giving only even numbers, as they will divide without fractions.

Think of	7
Double it	14
Add 10 to it	<u>10</u>
Halve it	2) <u>24</u>
Which will leave	<u>12</u>
Subtract the number thought of	<u>7</u>
<i>The remainder will be</i>	5

The Three Jealous Husbands.

Three jealous husbands, A, B and C, with their wives being ready to pass by night over a river, find at the water-side a boat which can carry but two at a time, and for want of a waterman they are compelled to row themselves over the river at several times. The question is, how those six persons shall pass, two at a time, so that none of the three wives may be found in the company of one or two men, unless her husband be present?

This may be effected in two or three ways; the following may be as good as any: Let A and wife go over—let A return—let B's and C's wives go over—A's wife returns—B and C go over—B and wife return, A and B go over—C's wife returns, and A's and B's wives go over—then C comes back for his wife. Simple as this question may appear, it is found in the works of Alcuin, who flourished a thousand years ago, hundreds of years before the art of printing was invented.

The Arithmetical Mouse-Trap.

One of the best and most simple mouse-traps in use may be constructed as follows: Get a slip of smooth pine, about the eighth of an inch thick, a quarter of an inch broad, and of sufficient length to cut out the following parts of a trap: First, an upright piece, three or four inches high, which must be square at the bottom, and a small piece to be cut off at the top to fit a notch in No. 2.

The second piece must be of the same length as the first, with the notch cut across nearly at the top of it, to fit the top of No. 1, and the other end of it trimmed to catch the notch in No. 3. The third piece should be twice as long as either of the others; a notch, similar to that in No. 2, must be cut in one

end of it to catch the lower end of No. 2. Having proceeded thus far, you must put the pieces together, in order to finish it, by adding another notch in No. 3, the exact situation of which you will discover as follows: Place No. 1 upright, then put the notch of No. 2 in the thinned top of No. 1; then get a flat piece of wood, or a slate, one end of which must rest on the ground, and the center of the edge of the other on the top of No. 2. You will now find the thinned end of No. 2 elevated by the weight of the flat piece of wood or slate; then put the thinned end of it in the notch of No. 3, and draw No. 2 down by it, until the whole forms a resemblance of a figure 4; at the exact place where No. 3 touches the upright, cut a notch, which, by catching the end of No. 1, will keep the trap together. You may now bait the end of No. 3 with pieces of cheese; a mouse, by nibbling the bait, will pull down No. 3, the other pieces immediately separate, and the slate or board falls upon the mouse. We have seen numbers of mice, rats and birds caught by this.

HOW TO BECOME A CHEMIST.

In the eleventh century, and during the reign of King Henry the First, surnamed Beauclerk, or the fine scholar, there appeared for the first time in certain books, professing to teach the art of making gold, the words chemistry, chemist, derived from the Greek. Seven hundred years and more have passed away, and that which was only the pursuit of a shadow called alchemy, has resulted in the acquisition of a great and noble science, now and again called chemistry. So it is with the great edifice Chemistry; we may, in these brief pages, peep in at the open door, but should we desire to go beyond the threshold, there are numerous guides, such as Roscoe, Wilson, and Fownes, who will conduct us through the mazes of the interior, and explain in elementary language the beautiful processes which have become so useful to mankind.

Chemistry is one of the most comprehensive of all the sciences, and at the same time one which comes home to us in the most ordinary of our daily avocations. Most of the arts of life are indebted to it for their very existence, and nearly all have been, from time to time, improved by the application of its principles.

Chemistry is, in fact, the science which treats of the composition of all material bodies, and of the means of forming them into new combinations, and reducing them to their *ultimate elements*, as they are termed; that is, bodies which we are unable to split up, as it were, or separate into other bodies. To take a common substance as an illustration; water, by a great number of processes, can be separated into two other substances, called oxygen and hydrogen, in the proportion by weight of 8 parts of the first to 1 of the second; but no power that we at present possess can separate the oxygen and hydrogen into any other bodies; they are therefore called ultimate elements, or undecomposable bodies.

Again, sulphate of magnesia (common Epsom salts) can be very easily separated into two other substances,—sulphuric acid and magnesia; and in this instance, both these substances can again be subdivided—the acid into sulphur and oxygen, and the magnesia into a metallic body called

magnesium and oxygen; but sulphur, oxygen, and magnesium are incapable of further division, and are therefore called *ultimate elements*.

These ultimate elements amount to 64 in number, according to the present state of our knowledge, and may be arranged in various ways; the simplest plan, perhaps, is dividing them into Non-metallic and Metallic elements.

The Non-metallic elements are:—1. Oxygen. 2. Hydrogen. 3. Nitrogen. 4. Chlorine. 5. Iodine. 6. Bromine. 7. Fluorine. 8. Carbon. 9. Sulphur. 10. Selenium. 11. Tellurium. 12. Silicon. 13. Boron. 14. Phosphorus. The last-named element is the connecting link with the metals through arsenic, which phosphorus closely resembles in its chemical properties.

The Metallic elements may be sub-divided into the metals of the alkalies, the metals of the alkaline earths, the metals of the earths, and the other metals sometimes called metals proper.

1st. The metallic bases of the alkalies:—potassium, sodium, lithium, ammonium, cæsium, rubidium.

2d. The metallic bases of the alkaline earths:—calcium, strontium, barium.

3d. The metallic bases of the earths:—aluminum, glucinum, zirconium, thorium, yttrium, erbium, cerium, lanthanum, didymium.

4th. The metals proper, the most important of which are:—platinum, gold, silver, mercury, copper, iron, tin, lead, nickel, zinc, bismuth, antimony, manganese, cobalt, arsenic.

Now, from these elementary bodies, united together in various proportions, is formed the infinite variety of substances around us, whether animal, vegetable, or mineral; in fact, a few only are generally employed:—in the case of animals and vegetables, oxygen, hydrogen, carbon, nitrogen, with occasionally some sulphur, calcium, phosphorus, and silicon, suffice for building up the beautiful forms of animated nature; while the fabric of our globe itself consists for the most part of the earths; silex, *i.e.*, flint or crystal; lime, in the shape of chalk, marble, or limestone, such as our flagstones are composed of; slate and granite, which are compounds of aluminium, silica, and small quantities of oxide of iron, and sometimes a little potash, etc.; and through their masses are projected irregular streams—veins as they are termed—of the metals, either in a pure state, as is the case sometimes with gold, silver, platinum, mercury, and perhaps one or

two others; or combined with one of the non-metallic elements, or with one another.

Late calculations have determined the composition of the earth's solid crust in 100 parts by weight to be:

Oxygen	44.0	to	48.7.
Silicon	22.8	“	36.2.
Aluminium	9.9	“	6.1.
Iron	9.9	“	2.4.
Calcium	6.6	“	0.9.
Magnesium	2.7	“	0.1.
Sodium	2.4	“	2.5.
Potassium	1.7	“	3.1.
	100.0		100.0

All these combinations are effected by certain powers, termed *forces*; those which cause the union of the elements are called the forces of attraction; those causing their separation, the forces of repulsion.

The force of attraction when exerted between masses of matter, is termed gravitation; when it unites particles of matter of a similar kind and produces masses, it is called the attraction of cohesion; when the particles united are of a dissimilar character, it is then termed chemical or elective affinity. For example, the crystals of Epsom salts are formed from minute particles of the salt, united into a larger or smaller mass by the attraction of cohesion, while the *elements* of which each particle consists, namely, the sulphur, oxygen, and magnesium, are united by the attraction of chemical affinity.

Cohesion thus unites particles of a similar kind; chemical affinity, of a dissimilar nature. It is to cohesion that the existence of *masses* of matter is owing, and its power increases as the squares of the distances diminish, in an inverse ratio to the squares of the distances of the particles on which it acts.

The power exerted by cohesion may be exhibited in various ways. This is one: Procure two discs of glass about three inches in diameter, their surfaces being ground extremely smooth; fix each into a square piece of wood, taking care that they are placed accurately in the center; then put them together by sliding their edges very carefully over each other, so as to avoid

any air getting between them, and you will find a great force necessary to separate them. A hook should be fixed into the center of each piece of wood, so that they may be suspended, and a weight hung to the lower one. It is almost impossible for any one to separate them by merely pulling them with both hands; a weight of many pounds is required for that purpose. In like manner two freshly-cut surfaces of caoutchouc will, on being squeezed together, cohere so perfectly, that it is difficult to tear them asunder, and it is in this way that tubes of caoutchouc may be rapidly prepared for experiments, where little or no pressure is exerted.

Chemical affinity is sometimes called *elective*, or the effect of *choice*, as if one substance exerted a kind of *preference* for another, and chose to be united to it rather than to that with which it was previously combined; thus, if you pour some vinegar, which is a weak acetic acid, upon some pearlash (a combination of potash and carbonic acid), or some carbonate of soda (a combination of the same acid with soda), a violent effervescence will take place, occasioned by the escape of the carbonic acid, displaced in consequence of the potash or soda preferring the acetic acid, and forming a compound called an acetate. Then if some sulphuric acid be poured on this new compound, the acetic acid will in its turn be displaced by the greater attachment of either of the bases, as they are termed, for the sulphuric acid. Again, if into a solution of blue vitriol (a combination of sulphuric acid with oxide of copper) the bright blade of a knife be introduced, the knife will speedily be covered with a coat of copper, deposited in consequence of the acid *preferring* the iron, of which the knife is made, a quantity of it being dissolved in exact proportion to the quantity of copper deposited.

It is on the same principle that a very beautiful preparation called a silver-tree, or a lead-tree, may be formed thus:—Fill a wide bottle, capable of holding from half a pint to a pint, with a tolerably strong solution of nitrate of silver (lunar caustic), or acetate of lead, in pure distilled water; then attach a small piece of zinc by a string to the cork or stopper of the bottle, so that the zinc shall hang about the middle of the bottle, and set it by where it may be quite undisturbed; in a short time, brilliant plates of silver or lead, as the case may be, will be seen to collect around the piece of zinc, assuming more or less of the crystalline form. This at first is a case of elective affinity; the acid with which the silver or lead was united *prefers* the zinc to either of those metals, and in consequence discards them in order

to attach the zinc to itself, subsequently a voltaic current is set up between the two metals, and the process will continue until almost the whole of the zinc is taken up, or nearly the whole of the silver or lead deposited.

Again, many animal and vegetable substances consist for the most part of carbon or charcoal, united with oxygen and hydrogen in the proportion which forms water. Now oil of vitriol (strong sulphuric acid) has so powerful an affinity, or so great a *thirst* for water, that it will abstract it from almost any body in which it exists; if you then pour some of this acid on a lump of sugar, or place a chip of wood in it, the sugar or wood will speedily become quite black, or be *charred*, as it is called, in consequence of the oxygen and hydrogen being removed by the sulphuric acid, and only the carbon, or charcoal, left.

When Cleopatra dissolved pearls of wondrous value in vinegar, she was exhibiting unwittingly an instance of chemical elective affinity; the pearl being simply carbonate of lime, which was decomposed by the greater affinity or fondness of lime for its new acquaintance (the acetic acid of the vinegar) than for the carbonic acid, with which it had been united all its life,—an example of inconstancy in strong contrast with the conduct of its owner, who chose death rather than become the mistress of her lover's conqueror.

Gases.

The three permanent gaseous elements are oxygen, hydrogen, and nitrogen.

The compound gases are very numerous, some being combustible, and others supporters of combustion.

Gases are for the most part transparent and colorless, with a few exceptions, and of course, like the air of the atmosphere, invisible. They are little affected by the attraction of cohesion, but rather, on the contrary, the particles composing them have a constant tendency to separate from each other, so that their force of expansion is only limited by the pressure under which they may be kept, and the temperature they may be exposed to. They have a tendency to *penetrate* each other, as it were; for instance, if you take a jar of heavy gas, such as carbonic gas, set it with its mouth upwards, then invert over it another jar containing hydrogen, a gas nearly twenty-two times lighter, in a very short time the two gases will have become

thoroughly mixed, the heavy carbonic acid having risen, and the light hydrogen fallen, until the gases are thoroughly mixed, each jar containing an equal quantity of each gas.

Oxygen Gas.

This gas, so named from two Greek words signifying the maker of acid, was discovered by Dr. Priestly in 1774. He obtained it by heating the red oxide of mercury in a glass retort, when the gas escaped in considerable quantities. In the ensuing year Scheele obtained it by a variety of methods, and a few years afterwards Lavoisier discovered that it was contained in atmospheric air, where it exists in the proportion of about one-fifth, the remaining four fifths being almost entirely nitrogen.

Oxygen gas may be obtained for the purpose of experiment, by heating to redness the black oxide of manganese in an iron bottle, to the mouth of which a flexible tube is attached to convey away the gas as fast as it is liberated from the manganese. The first portions should be allowed to escape, being mixed with the air in the tubes and bottle, and the remainder may be collected in a gasometer, or in glass jars inverted over water.

Another method to obtain the gas, and one to be used only in the absence of other ingredients, is to mix in a retort some of this same oxide of manganese with about half its weight of strong sulphuric acid, and apply heat to the retort, when the gas will come over in considerable quantities; the first portions must be allowed to escape as before.^[1] If the gas is required very pure, a small quantity of the salt called chlorate of potassa, may be heated in a retort, and oxygen gas will be evolved, and may be collected as before. If you have an iron bottle, the first mode is by far the cheapest, as the heat of a bright fire is sufficient for the operation, and a large quantity of gas is obtained in a short time from a very inexpensive material. The most rapid and convenient process of all is to heat a mixture of two parts chlorate of potash, and one of powdered black oxide of manganese, in a common clean oil flask, to which a cork and bent tube has been adapted. Care must be taken not to mistake sulphide of antimony for black oxide of manganese, as very serious accidents have arisen from this cause.

Oxygen is largely distributed over our globe, both in its uncombined state, and in union with other substances. Besides forming one-fifth of the atmosphere, it forms eight-ninths by weight of all the water in the ocean, rivers, and springs on the face of the whole earth. It also, in combination with various metals, forms the various earths and minerals of which the crust of the earth consists, so that it is the most abundant and widely distributed substance in nature, and in combination with other elements, forms nearly half the weight of the solid earth.

In its uncombined state it is a colorless gas, somewhat heavier than atmospheric air, without taste or smell. It is a powerful supporter of combustion, and is absolutely necessary for the support of animal life, which cannot exist for any time without a free supply of this gas, which is constantly consumed in the act of breathing and is replaced by an equivalent portion of carbonic acid gas. The want of oxygen is partly the cause of the oppression felt in crowded rooms, where the air cannot be renewed so fast as is required for the number of persons who are constantly consuming the oxygen; and if an animal be confined under a glass jar inverted over water, it will presently die, just for the same reason that burning tapers are extinguished under similar circumstances.

If a jet of this gas be thrown upon a piece of charcoal, sulphur, or almost any combustible body in a state of ignition, it will make it burn with great vividness and rapidity.

FOOTNOTE:

[1] Some *boiling* water should be added to the mass left in the retort directly the gas has ceased to come away, or it will adhere to the glass so firmly, that the retort will certainly be spoilt.

Experiment.

But by far the most intense heat, and most brilliant light, may be produced by introducing a piece of phosphorus into a jar of oxygen. The phosphorus may be placed in a small copper cup, with a long handle of thick wire passing through a hole in a cork that fits the jar. The phosphorus must first be ignited; and, as soon as it is introduced into the oxygen, it gives out a light so brilliant that no eye can bear it, and the whole jar appears filled

with an intensely luminous atmosphere. It is well to dilute the oxygen with about one-fourth part of common air to moderate the intense heat which is nearly certain to break the jar if pure oxygen is used.

Experiment.

If a piece of charcoal, which is pure carbon or nearly so, be ignited, and introduced into a jar containing oxygen or common atmospheric air, the product will be carbonic gas only, of which we shall speak presently. As most combustible bodies contain both carbon and hydrogen, the result of their combination is carbonic acid and water. This is the case with the gas used for illumination; and in order to prevent the water so produced from spoiling goods in shops, various plans have been devised for carrying off the water when in the state of steam. This is generally accomplished by suspending over the burners glass bells, communicating with tubes opening into the chimney, or passing outside the house.

To show that oxygen, or some equivalent, is necessary for the support of combustion, fix two or three pieces of wax-taper on flat pieces of cork, and set them floating on water in a soup-plate, light them, and invert over them a glass jar; as they burn, the heat produced may perhaps at first expand the air so as to force a small quantity out of the jar, but the water will soon rise in the jar, and continue to do so until the tapers expire, when you will find that a considerable portion of the air has disappeared, and what remains will no longer support flame; that is, the oxygen has been converted partly into water, and partly into carbonic acid gas, by uniting with the carbon and hydrogen, of which the taper consists, and the remaining air is principally nitrogen, with some carbonic acid; the presence of the latter may be proved by decanting some of the remaining air into a bottle, and then shaking some lime-water with it, which will absorb the carbonic acid and form chalk, rendering the water quite turbid.

Nitrogen.

This gas is, as its name implies, the producer of niter, or at least forms a portion of the nitric acid contained in niter. It is rather lighter than atmospheric air, colorless, transparent, incapable of supporting animal life, on which account it is sometimes called azote—an objectionable name, as it

is not a poison like many other gases, but destroys life only in the absence of oxygen. This gas extinguishes all burning bodies plunged into it, and does not itself burn. It exists largely in nature, for four-fifths of the atmosphere consists of nitrogen gas. It is also an important constituent of animal bodies, and is found in the vegetable world.

Nitrogen may be most easily obtained for experiment by setting fire to some phosphorus contained in a porcelain or metallic cup, placed under a gas jar full of air, and resting on the shelf of the pneumatic trough, or in a soup-plate filled with water.

Nitrogen combines in five different proportions with oxygen producing five distinct chemical compounds, named respectively nitrous oxide, nitric oxide, nitric tri-oxide, nitric tetroxide, nitric pent-oxide, which last, united with water, forms nitric acid, now called hydric nitrate, as nitrous acid is termed hydric nitrite.

Nitrous oxide gas is generally known by the name of “laughing gas,” from the jolly sensations experienced on inhaling it. It may be procured by distilling in a glass retort a salt called nitrate of ammonia, which yields the gas in considerable quantities, and it should be kept standing in jars over water for some hours before it is used. It should be transferred into a silk air-tight bag, furnished with a stop-cock and mouth-piece, from which the gas may be breathed; a little practice is required to do this easily, and more resolution to desist when the gas begins to produce its effects, as it appears to fascinate the experimenter, and actual force is often necessary to remove the bag from the mouth. The effects produced vary according to the temperament of the person inhaling it; they are, however, always of a highly pleasurable nature, muscular action being generally greatly exalted, compelling the individual to race round the apartment and execute leaps and pirouettes perfectly astounding. Some persons shout and sing, and I have seen one expend his superfluous animation in twisting his features into such ludicrous grimaces as would be the envy of the candidates at a grinning match, and beat them all out of the field.

This gas is heavier than air, and supports combustion nearly as energetically as oxygen, as may be shown by introducing a piece of ignited phosphorus into a jar of this gas. It will not, however, support the life of small animals, such as mice, which introduced into it die very quickly.

The next compound of nitrogen with oxygen, when one proportion of nitrogen unites with two of oxygen, is termed *nitric oxide* gas. It may be easily procured by heating in a retort some copper turnings in dilute nitric acid. It is colorless and transparent, and has the property of combining with oxygen to form other compounds.

Experiment.

Into a jar of this gas standing over water pass some oxygen gas. The jar will be filled with red fumes, which will be rapidly absorbed by the water. If atmospheric air be used instead of oxygen, there will remain in the jar the nitrogen of the air amounting to four-fifths of the air employed.

This gas is destructive to animal life, in consequence of its property of uniting with the oxygen in the lungs, and producing the highly corrosive nitrous acid gas. It will, however, support the combustion of a few substances, phosphorus for instance, provided it is sufficiently heated before being plunged into the gas.

We pass over the third and fourth compounds of nitrogen with oxygen, as they are not calculated for amusing experiments. Nitric acid is easily prepared on the small scale, by gradually heating equal parts by weight of nitric and sulphuric acid in a retort to which a receiver has been adapted. The receiver, which may be a clean oil flask, should be kept cool with wetted blotting paper.

Nitrogen combines with chlorine and iodine, forming detonating compounds, the former being so extremely dangerous that it will be better to pass it by.

The compound with iodine, called iodide of nitrogen, may very easily be made by pouring a strong solution of ammonia (a compound of nitrogen and hydrogen) upon some iodine in a vial, shaking them well together, and after letting them stand for a few hours, pouring off the fluid; the black powder remaining in the vial is the explosive compound, the iodide of nitrogen. When dry, it is very apt to detonate spontaneously; it should therefore be shaken out of the vial while *wet*, and spread in very small quantities on separate pieces of blotting paper, which should be kept apart from each other. When thoroughly dry, the slightest touch with the point of a feather, shaking the paper on which it rests, or even opening too rapidly

the door of a closet where it has been put to dry, will cause it to explode, producing a quantity of violet-colored fumes. The explosion is somewhat violent, producing a sharp cracking noise; and the greatest care should be taken in experimenting with it.

Atmospheric Air.

As has been already mentioned, nitrogen is the principal constituent of the air of the atmosphere which surrounds our globe, extending to a height of about forty-five miles above it, and playing a most important part in the economy of nature, inorganic as well as organic.

This atmospheric air consists by volume of nearly four-fifths of nitrogen, and rather more than one-fifth of oxygen, viz.: seventy-nine of the former to twenty-one of the latter, or twenty-three parts by weight of oxygen and seventy-seven of nitrogen; it generally contains also a variable proportion of the vapor of water, and a very small quantity of carbonic acid gas, being only about four volumes to 10,000 of air. Its constituent parts are easily separated, as it is a mechanical mixture and not a chemical compound, though the mixture by diffusion is so complete that chemists have not been able to ascertain any difference in the composition of air taken from all parts of the world, and from different heights, up to the highest point which has to this time been attained.

The atmosphere presses on the surface of the globe, and every being on it, with a force of about fifteen pounds to every square inch of surface, but as it presses equally in all directions, upwards as well as downwards, its weight cannot be perceived unless the pressure be removed from one surface by some artificial means.

Atmospheric air contains, besides the oxygen and nitrogen, its principal constituents, a small proportion of carbonic acid gas, as has been mentioned, and this may be shown by filling a tube about half full of lime-water, and shaking it with the air contained in the other half, when it will become slightly turbid from the insoluble carbonate of lime formed.

When we consider that every living animal is constantly consuming oxygen, and replacing it by carbonic acid gas, and that all burning bodies, fires in our dwellings, furnaces, artificial lights of all kinds, act in the same way in abstracting the oxygen from the air, and replacing it by immense

quantities of carbonic acid gas, which is a poison to all animals who breathe, or attempt to breathe it, we must wonder what becomes of this irrespirable gas, as it is found to exist in the air in quantities so minute, and by what means the oxygen is restored, and the air again made fit for respiration. This is effected by one of those laws which the wisdom of the Creator has impressed upon matter, by which one part of creation as it were balances another, and all proceeds in an endless circle of change. This carbonic acid, which is so poisonous to animal life, is the food of the vegetable world, plants having the power of taking up the carbonic acid into their pores; converting the carbon into their own substance, and rejecting the oxygen, which is again respired by animals, &c. In the same way, all animal refuse is the food of vegetables, and is used under the name of manures.

The atmosphere contains also a variable quantity of vapor of water, invisible as long as it is in the state of vapor, but it may be rendered obvious by bringing any very cold body into warm air, when the vapor will condense on the cold body in the form of small drops of water. A tumbler of fresh-pumped water brought into a crowded room is almost immediately covered with moisture, and it may also be seen on bottles of wine which have been put into ice before coming to table. Fogs are occasioned by the condensation of vapor produced by mixing a current of warm air with a colder air. The banks of Newfoundland are notorious for dense fogs, occasioned by the warm air brought from the south by the great Gulf Stream, mixing with the cold air from the Arctic regions, and thus precipitating the vapor in a visible form, rendering everything but itself invisible. The famous London fogs depend upon the same precipitation of the vapor of water, with the addition of the smoke from the numerous sea-coal fires, which give it that interesting yellow tinge for which it is so remarkable.

Aqueous vapor appears to impart a transparency to air, and permits objects to be seen more distinctly in proportion to its quantity; hence, when distant hills appear nearer, and objects upon them more distinct than usual, rain may be expected, the air being fully charged with vapor ready to be deposited on the slightest cause.

Hydrogen.

Hydrogen gas is the lightest substance known, being fifteen times lighter than atmospheric air. It is colorless and transparent, incapable of supporting combustion or respiration, but is itself combustible. Hydrogen, as its name implies (being derived from two Greek words, signifying the generator of water), is a constituent of water in the proportion of one-ninth by weight, and is always obtained by decomposing that fluid, by presenting to it some body to take up its other ingredient, oxygen, and so set the hydrogen at liberty. If the steam of water be passed through a red-hot gun-barrel, containing iron filings, the water is decomposed, the iron taking the oxygen, and the hydrogen comes over in torrents; but as every one has not a gun-barrel and furnace to heat it, the usual mode is to employ dilute sulphuric acid, and iron filings, or zinc, in small pieces, and it may be collected over water by means of a bent tube issuing from the bottle in which it is formed. It is so light that it was used to fill balloons before coal gas was to be had, and if you procure a light air-tight bag of silk, or thin membrane, such as a turkey's crop, and fill it with gas, it will ascend rapidly, and dance about the ceiling of a room.

Experiments.

1. Attach a tobacco-pipe to a bladder filled with this gas, and blow some soap-bubbles with it; they will rise very rapidly, and if a lighted taper be applied to them they burn.

If you mix in a soda-water bottle one-third of oxygen with two-thirds of hydrogen, and apply flame, the mixture will explode with a sharp report. Great care must be taken in all experiments with the mixed gases. To avoid danger the gases are placed in separate India-rubber bags, and are only brought together at the jet. This is an expensive apparatus, and should only be used by experienced persons.

2. If a jar of this gas be held with its mouth *downwards*, and a lighted taper passed up well into the jar, the taper will be extinguished, and the gas take fire, and burn quietly at the mouth of the jar; if mixed with oxygen or atmospheric air, it will explode.

Hold open the jet of hydrogen issuing from a small tube, hollow cylinders of glass or earthenware, Florence flasks, or hollow glass balls, and musical sounds will be produced, which were supposed to depend on some peculiar

property of hydrogen gas, until Mr. Faraday tried flame from coal gas, olefiant gas, and even the vapor of ether, when the sounds were still produced, and he attributed them to a continuous explosion, or series of explosions, produced by the union of oxygen with the hydrogen of the flames.

Water.

With oxygen, hydrogen unites to form the important compound water, which exists not only in the obvious form of oceans, rivers, lakes, rains, dews, &c., &c., but is found intimately combined with many substances, giving them some of their peculiar properties. Many crystals have a definite proportion of water combined with them, and on losing this water they lose their crystalline form. Many acids also cannot exist as acids without water. The slaking of lime depends upon the union of water with the lime, the dry powder resulting from the process being a *hydrate* of lime, the water having become *solidified*, and in passing from the fluid to the solid state gives out its latent caloric, producing the heat observed during the process. When a large quantity of lime, a barge-load for instance, has got wetted by accident, the heat evolved has been sufficient to set fire to the barge.

At the temperature of 32° of Fahrenheit's thermometer, water loses its fluid form, and becomes ice. As it solidifies, it starts into beautiful crystals, which unite and cross each other at determinate angles. Ice is lighter than the water on which it floats, forming a protection to the water beneath, and preventing it from being frozen so rapidly; else, if the ice were *heavier* than water, and consequently sank as soon as formed, each portion of water would be frozen in its turn, until rivers became solid throughout, and every living creature in them must be destroyed. Now, the temperature of the water under the ice is seldom much below 40°, and if care be taken to break holes at intervals to allow access to the air, the fish and other aquatic animals seldom suffer even in our coldest winters.

Although it is impossible to raise ice even one degree above 32° without thawing, it is not difficult to reduce water many degrees below that point without freezing it.

In order to obtain both the constituents of water in a separate state, it must be decomposed by galvanism, each pole of a battery terminating in a

separate tube containing water, when the result will be that at the positive pole oxygen gas will be evolved, and hydrogen at the negative, the latter being double the quantity of the former. Now, if you mix the gases thus obtained, introduce them into a vessel called a “Eudiometer,” and pass an electric spark through them from a Leyden vial, a sudden flash will be seen, and the gases will entirely disappear, being again converted into water. If you have a mercurial trough, and perform this experiment over mercury, the inside of the eudiometer will exhibit minute drops of water. Thus you have proved both by *analysis* and *synthesis*, that water consists of oxygen and hydrogen, in the proportion of one volume of the former to two of the latter.

Experiment.

Take some perfectly pure distilled water, filter it, surround it with a mixture of light snow, or powdered ice, and salt, taking care to keep it perfectly still, a thermometer having been previously placed in it. The mercury will gradually sink many degrees below the freezing point 32° (it has been reduced as low as 4°), the water still remaining fluid; when all at once, either from shaking the table, or simply because the reduction can be carried no further, it suddenly starts into ice, and the thermometer jumps up at once to 32° , where it remains until the whole is frozen, when the temperature gradually sinks to that of the surrounding medium.

Now if you remove the glass of ice from the freezing mixture into the apartment, and watch the thermometer, you will find it gradually rise to 32° , and there remain until all the ice is melted, when it will gradually acquire the temperature of the room. The reason of this is, that the water in passing from the solid to the fluid form absorbs, and in passing from the fluid to the solid form gives out caloric, so maintaining the temperature at 32° , the point at which the change of form takes place, until it is completed.

Between the temperature of 32° and 212° , water exists in a fluid form, under ordinary circumstances; but at the latter point it assumes the form of vapor or steam, and acquires many of the properties of gases, being indefinitely expansible by heat, the force increasing as the temperature is raised, provided the steam be confined, until it becomes irresistible—witness the frequent explosions of steam-engines in this country, where the engines are worked at a high pressure.

The temperature at which water boils is modified by the pressure applied to it. Thus, as you ascend a mountain, and so pass through a portion of the atmosphere, water boils at a lower temperature, until at great heights it boils at so low a heat, that good tea cannot be made because it is impossible to heat the water sufficiently. Under the exhausted receiver of an air-pump, water boils at about 140°.

Chlorine.

Another gaseous element, sometimes called a supporter of combustion, is named chlorine, from a Greek word signifying yellowish green.

This gas was formerly called “oxymuriatic acid,” being supposed to be a compound of oxygen and muriatic acid gases, until Sir H. Davy, in a series of masterly experiments carried on during the years 1808-9-10 and 11, proved that it contained no oxygen or muriatic acid, and that it was in fact a simple or undecomposed substance, and changed its name to chlorine, which name was, after some discussion, accepted by the scientific world, and is still in use.

This gas may be obtained for experiment, by gently heating in a retort a mixture of muriatic or hydrochloric acid, hydrochloride, as it is now called, with some black oxide of manganese: the muriatic acid, a compound of chlorine and hydrogen, is decomposed, and so is the oxide of manganese, giving out some of its oxygen, which takes the hydrogen from the muriatic acid to form water, while the chlorine gas, with which the hydrogen had been united, is set at liberty, and may be collected in jars over water.

Chlorine gas is transparent, of a greenish yellow color, has a peculiar disagreeable taste and smell, and if breathed even in small quantities, occasions a sensation of suffocation, of tightness in the chest, and violent coughing, attended with great prostration. I have been compelled to retire to bed from having upset a bottle containing some of this gas. It destroys most vegetable colors when moist, and is in fact the agent now universally employed for bleaching purposes.

It has also the power of combining with and destroying all noxious smells, and is invaluable as a purifier of foul rooms, and destroyer of infection. For these latter purposes it is used in combination with lime, either in substance or solution, under the name of “Chloride of Lime.”

Chlorine gas is a powerful supporter of combustion, many of the metals taking fire spontaneously when introduced in a fine state of division into the gas.

Experiments.

1. Into a jar of chlorine gas introduce a few sheets of copper leaf, sold under the name of Dutch foil, when it will burn with a dull red light.
2. If some metallic antimony in a state of powder be poured into a jar of this gas, it will take fire as it falls, and burn with a bright white light.
3. A small piece of the metal potassium may be introduced, and will also take fire.
4. A piece of phosphorus will also generally take fire spontaneously when introduced into this gas. In all these cases direct compounds of the substances with chlorine are produced, called chlorides.
5. If a lighted taper be plunged quickly into the gas, it will continue to burn with a dull light, giving off a very large quantity of smoke, being in fact the carbon of the wax taper, with which the chlorine does not unite; while the other constituent of the taper, the hydrogen, forms muriatic acid by union with the chlorine.
6. This substance has the property of destroying most vegetable colors and is used in large quantities for bleaching calico, linen, and the rags of which paper is made. It is a curious fact that it shows this property only when water is present, for if a piece of colored cloth is introduced dry into a jar of the gas, also dry, no effect will be produced—wet the cloth, and reintroduce it, and in a very short time its color will be discharged.
7. Introduce a quantity of the infusion of the common red cabbage, which is of a beautiful blue color, into a jar of this gas, and it will instantly become nearly as pale as water, retaining a slight tinge of yellow. A solution of sulphate of indigo can always be obtained, and answers well for this experiment.

Muriatic Acid Gas, or Chloride.

With chlorine, hydrogen forms a compound called muriatic, or hydrochloric acid gas. It cannot easily be formed by the direct union of its elements, but is procured from some compound in which it exists ready formed. Common salt (chloride of sodium) is generally employed; and when acted on by strong sulphuric acid (or oil of vitriol), the gas is disengaged in abundance. It must be collected over mercury, for water absorbs it, forming the liquid muriatic, or hydrochloric acid.

A lighted taper plunged into this gas is instantly extinguished. It is very dangerous to animal life if respired. It has the property of destroying animal effluvia, and was once employed to purify the cathedral of Dijon, which was so filled with putrid emanations from the bodies buried in it, that it had been closed for some time. It perfectly succeeded, but it is so destructive to all metallic substances that it is not used now, for the chlorides of lime and zinc have since been discovered to act more effectually than the muriatic acid gas, without its inconvenience.

The compounds of hydrogen with iodine are passed over.

With nitrogen, hydrogen unites and forms one of the most extraordinary compounds in the whole range of chemistry,—the gas called ammonia. This is the only gas possessing what are called alkaline properties; *i.e.*, it changes the blue color of certain vegetables to green, yellow to deep brown, and unites with the acids to form neutral compounds, just as the other alkalies, potash and soda, which are oxides of metals. It may be procured in abundance by heating the hydrochlorate of ammonia, or sal ammoniac, as it is usually called, with quick-lime, which takes the hydrochloric acid, and sets free this remarkable gas. It must be received over mercury, as it is absorbed to almost any extent by water, forming the fluid sold as “spirits of hartshorn” in the shops.

This gas is colorless and transparent, lighter than atmospheric air, and will not support combustion; it has a very pungent but not disagreeable smell. Under certain circumstances it is combustible.

Experiments.

1. Take a bottle containing chlorine gas, and invert over its mouth another filled with ammoniacal gas; then if the bottles be held in the hand (guarded by a pair of gloves), and suddenly turned, so that the chlorine be uppermost,

the two gases will unite so rapidly that a white flame fills the bottles for an instant.

2. Substitute for the chlorine of the last experiment a bottle of carbonic or hydrochloric acid gas; in either case the gases disappear, and a light white powder settles on the sides of the bottles, being the carbonate or hydrochlorate of ammonia, according to the acid used.

Carbonate of ammonia is the substance sold for “smelling salts;” and the hydrochlorate, or muriate of ammonia, is the salt called “sal ammoniac,” whence the alkaline gas was first obtained, and from which it got its name of ammonia. The salt itself was so called, because it was formerly brought from the deserts near the ruins of the temple of Jupiter Ammon.

This salt is, as has been shown, a compound of muriatic acid gas and ammoniacal gas, containing therefore only *three* simple elements—hydrogen, chlorine, and nitrogen, all gases, and known only in the gaseous state, its symbol being NH_4C_2 ; yet they by union form a solid body, resembling in all essential qualities the salts of potash and soda, which are oxides of known metals. Moreover, if some mercury be placed in a solution of this salt, and subjected to the action of galvanism, the *negative* pole being applied to the mercury, and the positive to the sal ammoniac, the mercury presently loses its fluidity, increases greatly in size, and in fact presents the same appearance as when it is mixed with some metal, forming what is called an “amalgam.” When the battery ceases to act, a succession of white films forms on the surface of the amalgam, and the mercury soon returns to its original state. How is this to be explained? Some chemists have supposed that there must be a *base* united to the mercury, and have named this hypothetical substance “ammonium,” to correspond to potassium and sodium, the bases of potash and soda, which resemble ammonia in so many properties. But what is this ammonium? and how is it formed? for hydrogen and nitrogen are simply elementary bodies. Are *all* metals compounds of gases? and are there but a few elements instead of the 64 now enumerated? This, however, is a difficult question, not fitted for discussion here.

Carbonate of ammonia may be obtained by mixing together powdered chalk (which is a carbonate of lime) and muriate of ammonia, and heating the mixture in close vessels, when the salt in question will rise in fumes, and be

condensed in a mass in the upper part of the vessel. It is, however, so largely produced in other manufactures, particularly in gas-works, that there is no necessity to resort to the more expensive and direct method. It is the well-known “smelling salts.”

The only other salt of ammonia worth our notice here is the nitrate, from the destructive distillation of which is obtained the nitrous oxide, or laughing gas, already mentioned.

Iodine—Bromine—Fluorine.

On the coasts of certain islands belonging to the Duke of Argyll, vast quantities of sea-weed are occasionally torn up from their ocean beds and deposited on the shores. This weed, after being partially dried by exposure to the sun and air, is burnt in a shallow pit; the ashes are then collected, and form the commercial raw material called kelp, from which iodine is procured by a gradual series of processes.

Experiments.

Iodine has a beautiful metallic luster, with a bluish black color, and should be kept in a well-stoppered bottle. A small quantity placed in a clear flask and heated, affords a magnificent violet vapor, which may be poured from the flask into another glass vessel, when it condenses again into crystalline plates. The color of the vapor originates the name of this element, so called from a Greek word, meaning violet-colored. If a little iodine be placed in contact with a thin slice of phosphorus, the latter takes fire almost immediately.

Bromine.

From the Greek, signifies a bad odor, and is most intimately allied with chlorine and iodine; like these elements, it belongs to the sea, and is a constituent of sea-water. Bromine is a very heavy fluid, and should be preserved by keeping it covered with water in a stoppered bottle.

Experiments with liquid bromine are not recommended, as all the most interesting ones can be performed with the vapor, which is easily procured by letting fall a few drops of bromine into a warm dry bottle.

Experiments.

Pounded antimony sprinkled into the vapor takes fire immediately.

A thin slice of phosphorus placed in a deflagrating ladle and placed into the vapor of bromine ignites very quickly.

A solution of sulphate of indigo, or an infusion of red cabbage, are easily bleached by being shaken violently with the vapor of bromine.

Fluorine.

In many parts of England, especially in Devonshire, Cornwall, and above all in Derbyshire, is found a very beautiful mineral, known by the name of Fluor Spar, Derbyshire Spar, and called by the miners Blue John, to distinguish it from another mineral found in the same locality, called *Black Jack*. It occurs in very regular and frequently large crystals in the form of cubes, and occasionally in octoëdra. It is a compound of calcium with fluorine, and is very abundant in certain fossil bones. This element, in combination with hydrogen and called hydrofluoric acid, acts so energetically upon all substances containing silica, that it cannot be preserved in vessels of glass or porcelain—very few of the metals are capable of resisting its action, lead being nearly the only common metal possessed of this power. Gutta percha may also be employed for vessels to hold it.

This property of dissolving silica has caused this acid to be used for engraving on glass.

Experiment.

Mix one part of fluor-spar, quite pure, with two parts of oil of vitriol, in a saucer, and apply a gentle heat, when the acid will be disengaged in the form of vapor. Prepare a piece of glass after the manner of engraving on copper, by coating it with a thin covering of wax, placing a paper over the wax, and then drawing any design with a sharp-pointed instrument, when, on removing the paper, the wax-coating will be found to be removed wherever the instrument has passed over it. Now invert this glass over the fumes of the acid for half an hour or so, and then heat the glass so as to soften the coating, and wipe it off; the design will then appear “bitten in,” as

the term is, that is, the acid will have dissolved the glass wherever it was not protected by the wax, and will exhibit the design indelibly fixed on the glass.

This acid requires the greatest care in handling, for it is extremely corrosive, producing very troublesome ulcers if it comes in contact with the skin; even the fumes will produce smarting if the skin is long exposed to them.

Carbon.

The next substance in our list of elementary bodies is named carbon.

The purest form of carbon is the precious stone called diamond, which consists entirely of carbon in a crystallized form. The French chemist Lavoisier was the first who proved the combustibility of the diamond; and Sir H. Davy found that when once set on fire it would continue to burn in oxygen gas air, and that the product of the combustion was carbonic acid gas, exactly equal in quantity to the gas produced by burning an equal weight of pure charcoal, the most common form of carbon.

Plumbago, or “black-lead,” as it is very improperly called, is also nearly pure carbon, a very small quantity of iron being united with it.

By far the greater part of all vegetable, and a very large portion of animal bodies consists of carbon; and in the state of carbonic acid in combination with lime and some other earths, it forms nearly the half of all the chalk, marble, and limestone of our hills; so that it is, in one shape or other, one of the most widely diffused bodies in nature.

Carbon forms two gaseous compounds with oxygen; the first, called carbonic oxide, is easily obtained by boiling oxalic acid with its own bulk of sulphuric acid, in a flask to which a cork and bent tube is attached. The gas comes over in large quantities, and must be collected in a gas jar, or the pneumatic trough. It is inflammable, and burns with a lambent blue flame.

The other compound, carbonic acid, is transparent, colorless, much heavier than atmospheric air, has an agreeable taste, has the power of irritating the mucous membrane of the nose (as any one can tell who has drunk soda-water), without possessing any particular odor, is absorbed by water, does not support respiration, and extinguishes flame.

Carbonic acid gas may be obtained with the greatest facility by pouring some muriatic or sulphuric acid, diluted with about six parts of water, upon some pieces of marble or limestone in a bottle with a tube attached, when the gas comes over in torrents. It may be collected over water.

Experiments.

1. To show the great comparative weight of this gas, place a lighted taper at the bottom of a tall glass jar, then take a jar full of carbonic acid gas, and pour it as you would pour water into the jar containing the lighted taper; you will soon find the taper will be extinguished as effectually as if you had poured water on it, and the smoke of the taper will float on the surface of the gas in very beautiful wavy forms.
2. Heat a piece of the metal potassium in a metal spoon (platinum is best), and if introduced in a state of ignition into the gas, it will continue burning brilliantly, producing a quantity of dense smoke, which is the carbon from the carbonic acid, the potassium having seized the oxygen, and being converted by it into potash.
3. If a mouse, bird, or other small animal be placed in a jar of this gas, it becomes insensible almost immediately; but if speedily removed it will occasionally recover.
4. Shake up some water with some of this gas in a bottle; the greater part of the gas will be absorbed by the water, which acquires a sparkling appearance and a pleasant sharp taste; with the addition of a little soda this becomes the well-known beverage called soda-water, so famous for removing the morning headaches caused by "*that salmon*" having disagreed at yesterday's dinner.

It is the presence of this gas which renders it so dangerous to descend into deep wells, for by its great weight it collects at the bottom, and instantly suffocates any unfortunate person who incautiously subjects himself to it. Hence it is prudent always to let down a lighted candle before any one descends into a well, or other deep excavation, and if the candle is extinguished, it is necessary to throw down several pails of water—lime-water, if possible—and again to try the candle, which must burn freely before it is safe for any one to descend.

It is this same gas, under the name of “choke-damp,” which proves so dangerous to miners, particularly after an explosion of “fire-damp,” for it is the principal product of the explosion, and it is by no means an easy matter to dislodge it.

Carbonic acid gas has been condensed into the fluid form by causing it to be disengaged under great pressure; the fluid acid has the appearance of water. When the pressure is removed, as by allowing some of the fluid acid to escape from the vessel in which it has been condensed, it instantly reassumes the gaseous form, and in so doing absorbs so much latent caloric that a portion of the acid is actually solidified, and appears in the shape of *snow*, which may be collected and preserved for a short time.

Carbonic acid and lime are mutually tests for each other. If a jar containing a little lime-water be put into a jar of this gas, it speedily becomes turbid, the gas uniting with the lime, and producing chalk (the carbonate of lime), which is insoluble in water.

This gas is produced in large quantities by the respiration of animals, as may be proved by respiring through a tube immersed in lime-water, when the water will be instantly rendered turbid from the formation of chalk.

Carbon and Hydrogen.

To the combination of these elements in various proportions, and with the occasional addition of other substances, we are indebted for all, or nearly all, our means of obtaining light and heat. Coal, wood, spirit, oil, and all the varieties of fats, are composed principally of carbon and hydrogen, and may easily be converted into the gas with which our houses and streets are lighted, which is nearly pure carbureted hydrogen.

The two chief definite gaseous compounds of those two elements are the light carbureted hydrogen and the heavy carbureted hydrogen, or olefiant gas. The first is easily procured by stirring the bottom of stagnant water on a hot summer’s day, and collecting the bubbles in a bottle filled with water and inverted over the place where the bubbles rise. This gas burns with a yellowish flame, and when mixed with a certain proportion of air, or oxygen gas, explodes with great violence on the application of a flame. It is the much dreaded fire-damp generated so profusely in some coal-mines,

and causing such fearful destruction to life and property when accidentally inflamed.

The other compound, the heavy carbureted hydrogen, forms part of the gas used for illumination; and, in fact, whatever substance is employed for artificial light, whether oil, tallow, wax, etc., etc., it is converted into this gas by heat, and then furnishes the light by its own combustion.

This gas has some very curious properties, and may be obtained nearly pure by mixing in a retort, *very carefully*, one part of spirits of wine and four of sulphuric acid. A lamp must be placed under the retort, when the gas will be speedily disengaged, and come over in great abundance; it may be collected over water.

This gas is transparent, colorless, will not support combustion, but is itself inflammable, burning with a brilliant white light, and being converted into carbonic acid and water. If mixed with three or four times its bulk of oxygen, or with common atmospheric air in much larger proportion, it explodes with great violence.

This gas is sometimes called “olefiant gas,” from the property it has of forming an oily substance when mixed with chlorine.

Experiment.

Into a jar standing over water half full of this gas, pass an equal quantity of chlorine gas. The gases will speedily unite and form an oily-looking liquid, which may be collected from the sides of the jar as it trickles down. By continually supplying the jar with the two gases as they combine, a considerable quantity of this substance may be collected. Care should be taken that the olefiant gas is rather in excess.

The substance produced is insoluble in water, with which it should be washed by shaking them together in a tube, and has a pleasant sweetish taste and aromatic smell, somewhat resembling ether.

Coal Gas.

The gas so universally employed for the purposes of illumination is a mixture of the carbureted and the bi-carbureted hydrogen, with minute portions of other gases scarcely worth mentioning. It is procured by

submitting coals to a red heat in iron retorts, having a tube passing from one end, along which passes all the fluid and gaseous matter separated from the coal, namely, gas tar, ammoniacal liquor, and various gases, carbureted hydrogen, carbonic acid, sulphureted hydrogen, etc., etc. The tar and ammoniacal liquor remain in the vessel in which the tubes from the retorts terminate, and the gaseous productions are conveyed through water and lime to separate the impurities; the remaining gas, now fit for use, passes into large iron vessels, called gasometers, inverted over water (like the jars in a pneumatic trough), whence it is sent through pipes and distributed where required. What remains in the retorts is called coke. It consists principally of charcoal mixed with the earthy and metallic particles contained in the coal.

Experiment.

If you possess an iron bottle, fill it with powdered coal, and attach a flexible tube to it, and put it in the fire; as soon as it becomes red-hot, large quantities of smoke will escape from the end of the tube, being the gas mixed with all its impurities. By passing it through water (if mixed with lime it will be better), the gas may be collected in jars standing over water, and submitted to experiment. If you do not possess a bottle, take a tobacco-pipe with a large bowl, fill the bowl with small coal, cover it with clay or putty, and when dry put it into the fire, and the gas will soon appear at the other end of the pipe, when it may be lighted, or the gas may be collected over water, as in the former experiment.

The light carbureted hydrogen contained in this gas is given off spontaneously in some coal-mines, and as it forms explosive mixtures with atmospheric air, the mines where it abounds could not be worked except at the greatest risk until about the beginning of the present century, when Sir H. Davy, while prosecuting some researches on the nature of flame, found that flame would not pass through metallic tubes, and he gradually reduced the length of the tubes, until he found fine iron-wire gauze formed an effectual barrier against the passage of flame. He then thought that if the light in a lantern were surrounded with this gauze it might safely be used in an inflammable atmosphere, where a naked light would instantly cause an explosion. Upon submitting the lamp to experiment, he found that by passing coal-gas by degrees into a vessel in which one of his lamps was

suspended, the flame first became much larger, and then was extinguished, the cylinder of gauze being filled with a pale flame, and though the gauze sometimes became red-hot, it did not ignite the gas outside. As the supply of coal-gas was diminished, the wick of the lamp was rekindled, and all went on as at first. A coil of platinum wire was afterwards suspended in the lamps, which becomes intensely heated by the burning gas, and gives out sufficient light to enable the miner to see to work. As long as the gauze is perfect it is almost impossible for the external air to be kindled by the wick of the lamp, but the miners are so careless that they will often remove the gauze to get a better light, to look for a tool, or some cause equally trivial, and many lives have been lost in consequence of such carelessness.

The effect of the fine wire gauze in preventing the passage of flame may be shown by bringing a piece of the gauze gradually over the flame of a spirit-lamp, until it nearly touches the wick, when the flame will be nearly extinguished, but the vapor of the spirit passes through, and may be lighted on the upper side of the gauze, which will thus have a flame on either side, though totally unconnected with each other. The flame from a gas-burner will answer as well as the spirit-lamp.

Nearly all the fluids, and solids also, used for procuring artificial light, such as naphtha, various oils, tallow, wax, spermaceti, spirits of wine, ether, etc., are compounds of carbon and hydrogen in different proportions, with the occasional addition of some other elements, especially oxygen and hydrogen, in the proportions to form water; as a general rule, those bodies containing the greatest proportion of carbon give the most light, though not necessarily the most heat.

Phosphorus.

The next body we have to notice is phosphorus, a most remarkable substance, procured from the earthy part of bones by a process not worth detailing here. It should be *always* kept under water, and the naked fingers should not be allowed even to touch it, for the smallest piece getting under the nail will inflame the first time the hand comes near the fire, and produce a sore very painful and difficult to heal. It should be cut under water by a knife or scissors, and removed with a pair of forceps. Its combustible properties have been frequently mentioned. It has also the property of shining in the dark, so that if you write on a wall with a solution of

phosphorus in oil, the letters will appear luminous in the dark—there is no danger, excepting from the greasiness of the oil.

Of the compounds of phosphorus with oxygen we have nothing to do here, but it forms with hydrogen a very curious gaseous compound, which takes fire spontaneously on the contact of air, or almost any gas containing oxygen.

Experiments.

It may be procured in either of two ways, according to the purpose for which it is wanted. The simplest way is to put a lump or two of phosphuret of lime into a saucer, about two inches in depth, containing some very diluted hydrochloric acid; bubbles of gas will speedily arise, and bursting on the surface of the fluid will burn with a slight explosion, and a circular wreath of smoke will rise into the atmosphere, enlarging as it rises, and wreathing itself round and round in the most elegant forms. Care must be taken that the phosphuret is *fresh*, and has been kept in a well-closed bottle, or the experiment will fail. The apartment must be free from draughts. If you desire to collect the gas, another method must be employed.

Fill a small retort *quite full*, neck and all, of a solution of caustic potash, drop five or six pieces of phosphorus into it, place the finger on the end of the retort, and immerse it in a basin also containing a *hot* solution of potash, remove the finger, and on applying the heat of a lamp to the retort, the gas will soon be disengaged rapidly, and drive out the fluid in the retort; it then escapes into the air, when it inflames with the same appearances as before described. Or it may be collected in gas jars filled with the potash solution, and held over the mouth of the retort. The object in using *hot* solution of potash in the basin is, that when the gas ceases to be given off, and the heat of the lamp is withdrawn, the hot fluid may gradually fill the vacuum which will form in the retort, and so prevent its being broken.

This gas is transparent and invisible, like most other gases. It is very poisonous if inhaled. If kept for any time, it loses its property of spontaneous inflammation, and must therefore be made at the time it is required.

Sulphur.

Sulphur, or brimstone, as it is frequently called, is sold in the form of sticks, or *roll* brimstone, or in fine powder called flowers of brimstone.

It is capable of showing electric phenomena when rubbed, giving out slight sparks, and first attracting and then repelling light bodies, such as small pieces of paper, etc. It is so bad a conductor of heat, that if grasped suddenly in a hot hand, it will crack and split into pieces just as glass does when suddenly heated or cooled—of course I am speaking of the *roll* brimstone. Water has no effect on it, as may be seen in the pans placed for pet dogs to drink out of, where the same piece of brimstone lies for years entirely unaltered, though it is supposed to prevent the dogs from having the mange!

Sulphur is largely used in the arts, principally in the manufacture of gunpowder, and fireworks of various kinds.

It combines with hydrogen and forms a gaseous compound called sulphureted hydrogen, which is almost the most poisonous of all the gases. It fortunately has so abominable a smell that due notice is given of its presence. Rotten eggs, a dirty gun-barrel, cabbage water, putrid animal and vegetable matter, etc., are indebted to this gas for their inviting odor; and it is found in certain mineral springs, as at Harrogate, where the water contains a considerable quantity of this gas, and is found useful in many diseases of the skin. It is also given off in a gaseous form by some volcanoes.

This gas may be obtained by pouring dilute hydrochloric acid upon a metallic sulphuret, such as that called crude antimony, being a native sulphuret of that metal. The gas may be kept for a short time over water. It is colorless and transparent, inflammable, but quite irrespirable, a small bird dying instantly when placed in air containing only 1-1500th of this gas. Its most remarkable property perhaps is the effect it has on certain metallic oxides and other metallic salts, blackening them instantly. White paint is easily stained by this gas, and it will darken the color of a metal in a solution, especially of lead, even when diluted with 20,000 times its weight of water. By way of experiment slips of ribbon, silk, or even paper, may be wetted with various metallic solutions, such as silver, mercury, lead, etc., or words may be written with the solutions, and on holding them over a stream of this gas they will be instantly darkened.

If this gas be collected in the pneumatic trough, which is usually painted *white*, you will have the pleasure of seeing the color changed to a very dark brown when your experiments are finished. With this very limited description of some of the non-metallic elements and their combinations, we must, for want of space, take leave of this division of chemistry, “the beginning of which is pleasure, its progress knowledge, its objects truth and utility.”

Metals.

We have a few words to say about a class of bodies called metals, which are of the utmost importance to mankind, and indeed without some of them, especially iron, few of the arts of civilized life could exist.

Fifty substances are now included in the list of metals; some of them, however, are only *supposed* to exist, such as *ammonium*, the supposed base of ammonia; and very many are to be viewed rather in the light of chemical curiosities, as from their great rarity they are too expensive for use, even if possessed of valuable properties of which others might be destitute.

Several metals have been known from the earliest period of which we have any record; such were iron, gold, silver, copper, lead, tin, mercury, and probably zinc, or at least its ores; for brass, which is an alloy of copper and zinc, is frequently mentioned in the early part of the Old Testament. In the sixteenth century others were discovered, such as antimony and bismuth. In the last century, cobalt, arsenic, platinum, nickel, manganese and chromium, together with several unimportant metals, were discovered by various philosophers; while in the present century, Dr. Wollaston discovered rhodium, the hardest and nearly the most indestructible of all the metals; and a few years later Sir Humphry Davy found that the alkalies, potash and soda, with many of the earths as they were called, had each a metal for its base, to which he gave the Latin name of the alkali or earth, with the termination *um*, as *potassium*, the base of potassa, *sodium* of soda, *calcium* of calx (lime), etc.

Until Sir H. Davy’s discovery of the metals of the alkalies, great specific gravity was regarded as one of the most striking characteristics of a metal, the lightest of them being much heavier than the heaviest earth; but potassium is very much lighter than water, and not much heavier than

spirits of wine. The other metals vary from a specific gravity of nearly twenty-one—or twenty-one times heavier than an equal bulk of water—that of platinum, to somewhat less than seven, which is the specific gravity of antimony.

When pure, they all have a luster, differing indeed among themselves, but so peculiar that it is called the metallic luster; for instance, gold and copper are yellow and red—nearly all the others white, but of a different shade; still there is no mistaking their metallic character, no other substances at all equaling them in this respect. They are also opaque, although some, like gold, when reduced to thin films, allow light to pass through them. They are all good conductors of heat and electricity, though some possess that property to a greater extent than others.

Many of them are what is called malleable, that is, may be extended or spread out by rolling, or beating them with a hammer; and ductile, or have the property of being drawn out into wire. Gold, silver, copper, and iron are the most remarkable in this respect.

All the metals are fusible, but some require very different degrees of heat to render them fluid—platinum requiring the heat of the oxy-hydrogen blowpipe, while tin melts in the flame of a candle, and mercury is fluid at all temperatures in this climate, but becomes solid at 40 degrees Fahrenheit below 0—a temperature occasionally experienced in the Arctic regions, where the mercurial thermometer is useless, the mercury becoming solid.

They are all excellent conductors of heat and electricity, and have the property of reflecting light and forming mirrors; for looking-glasses owe their power of reflecting objects principally to what is called the “silvering;” that is, a mixture of mercury and tin spread over the back of the glass, which being transparent, allows the image reflected from the metal to pass through it.

The following classification is most instructive, because it suggests to the young student that there must be identical properties in the metals thus placed together:

Class 1. Ammonium, cæsium, lithium, potassium, sodium.

Class 2. Calcium, barium, strontium.

Class 3. Aluminium, cerium, didymium, erbium, glucinium, lanthanum, thorium, yttrium, zirconium.

Class 4. Zinc class: cadmium, magnesium, zinc.

Class 5. Iron class: cobalt, chromium, indium, iron, manganese, nickel, uranium.

Class 6. Tin class: niobium, tantalum, tin, titanium.

Class 7. Tungsten class: molybdenum, tungsten, vanadium.

Class 8. Arsenic class: antimony, arsenic, bismuth.

Class 9. Lead class: lead, thallium.

Class 10. Silver class: copper, mercury, silver.

Class 11. Gold class: gold, iridium, osmium, palladium, platinum, rhodium, ruthenium.

Potassium.

Potassium was discovered by Sir H. Davy in the beginning of the present century. It is a brilliant white metal, so soft as to be easily cut with a penknife, and so light as to swim upon water, on which it acts with great energy, uniting with the oxygen, and liberating the hydrogen, which takes fire as it escapes.

Experiment.

Trace some continuous lines on paper with a camel's-hair brush dipped in water, and place a piece of potassium about the size of a pea on one of the lines, and it will follow the course of the pencil, taking fire as it runs, and burning with a purplish light. The paper will be found covered with a solution of ordinary potash. If turmeric paper be used, the course of the potassium will be marked with a deep brown color.—*Corollary.* Hence, if you touch potassium with *wet* fingers you will burn them.

If a small piece of the metal be placed on a piece of ice, it will instantly take fire, and form a deep hole, which will be found to contain a solution of potash.

In consequence of its great affinity for oxygen, potassium must be kept in some fluid destitute of that element, such as naphtha.

Caution!—As the globules of potassium after conversion into potash, when thrown on ice or water burst, strewing small particles of caustic hot potash in every direction, the greatest care should be taken to keep at a sufficient distance whilst performing the above experiment.

Saltpeter, or niter, is a compound of this metal (or rather its oxide) with nitric acid. It is one of the ingredients of gunpowder, and has the property of quickening the combustion of all combustible bodies.

Mix some chlorate of potash with lump sugar, both being powdered, and drop on the mixture a little strong sulphuric acid, and it will instantly burst into flame. This experiment also requires caution.

Want of space precludes us from considering the individual metals and their compounds in detail; it must suffice to describe some experiments showing some of their properties.

The different affinities of the metals for oxygen may be exhibited in various ways. The silver or zinc tree has already been described.

Experiments.

1. Into a solution of nitrate of silver in distilled water immerse a clean plate or slip of copper. The solution, which was colorless, will soon begin to assume a greenish tint, and the piece of copper will be covered with a coating of a light gray color, which is the silver formerly united to the nitric acid, which has been displaced by the greater affinity or *liking* of the oxygen and acid for the copper.

2. When the copper is no longer coated, but remains clean and bright when immersed in the fluid, all the silver has been deposited, and the glass now contains a solution of *copper*.

Place a piece of clean iron in the solution, and it will almost instantly be coated with a film of *copper*, and this will continue until the whole of that metal is removed, and its place filled by an equivalent quantity of *iron*, so that the nitrate of *iron* is found in the liquid. The oxygen and nitric acid

remain unaltered in quantity or quality during these changes, being merely transferred from one metal to another.

A piece of zinc will displace the iron in like manner, leaving a solution of nitrate of zinc.

Nearly all the colors used in the arts are produced by metals and their combinations; indeed, one is named *chromium*, from a Greek word signifying color, on account of the beautiful tints obtained from its various combinations with oxygen and the other metals. All the various tints of green, orange, yellow, and red, are obtained from this metal.

Solutions of most of the metallic salts give precipitates with solutions of alkalies and their salts, as well as with many other substances, such as what are usually called prussiate of potash, hydro-sulphuret of ammonia, etc.; and the colors differ according to the metal employed, and so small a quantity is required to produce the color that the solutions before mixing may be nearly colorless.

Experiments.

1. To a solution of sulphate of iron add a drop or two of a solution of prussiate of potash, and a blue color will be produced.
2. Substitute sulphate of copper for iron, and the color will be a rich brown.
3. Another blue, of quite a different tint, may be produced by letting a few drops or a solution of ammonia fall into one of sulphate of copper—a precipitate of a light blue falls down, which is dissolved by an additional quantity of the ammonia, and forms a transparent solution of the most splendid rich blue color.
4. Into a solution of sulphate of iron let fall a few drops of a strong infusion of galls, and the color will become a bluish-black—in fact, *ink*. A little *tea* will answer as well as the infusion of galls. This is the reason why certain stuffs formerly in general use for dressing-gowns for gentlemen were so objectionable; for as they were indebted to a salt of iron for their color, buff as it was called, a drop of tea accidentally spilt produced all the effect of a drop of ink.

5. Put into a largish test tube two or three small pieces of granulated zinc, fill it about one-third full of water, put in a few grains of iodine and boil the water, which will at first acquire a dark purple color, gradually fading as the iodine combines with the zinc. Add a little more iodine from time to time, until the zinc is nearly all dissolved. If a few drops of this solution be added to an equally colorless solution of corrosive sublimate (a salt of mercury) a precipitate will take place of a splendid scarlet color, brighter if possible than vermilion, which is also a preparation of mercury.

Crystallization of Metals.

Some of the metals assume certain definite forms in returning from the fluid to the solid state. Bismuth shows this property more readily than most others.

Experiment.

Melt a pound or two of bismuth in an iron ladle over the fire; remove it as soon as the whole is fluid; and when the surface has become solid break a hole in it, and pour out the still fluid metal from the interior; what remains will exhibit beautifully-formed crystals of a cubic shape.

Sulphur may be crystallized in the same manner, but its fumes, when heated, are so very unpleasant that few would wish to encounter them.

One of the most remarkable facts in chemistry, a science abounding in wonders, is the circumstance, that the mere contact of hydrogen, the *lightest* body known, with the metal platinum, the heaviest, when in a state of minute division, called spongy platinum, produces an intense heat, sufficient to inflame the hydrogen; of course this experiment must be made in the presence of atmospheric air or oxygen.

Time and space (or rather the want of them) compel us to conclude with a few experiments of a miscellaneous character.

To Form a Solid From Two Liquids.

Prepare separately, saturated solutions of sulphate of magnesia (Epsom salts) and carbonate of potash. On mixing them the result will be nearly solid.

Solutions of muriate of lime and carbonate of potash will answer as well.

To Form a Liquid From Two Solids.

Rub together in a Wedgewood mortar a small quantity of sulphate of soda and acetate of lead, and as they mix they will become liquid.

Carbonate of ammonia and sulphate of copper, previously reduced to powder separately, will also, when mixed, become liquid, and acquire a most splendid blue color.

The greater number of salts have a tendency to assume regular forms, or become *crystallized*, when passing from the fluid to the solid state; and the size and regularity of the crystals depend in a great measure on the slow or rapid escape of the fluid in which they were dissolved. Sugar is a capital example of this property; the ordinary loaf-sugar being rapidly boiled down, as it is called: while to make sugar-candy, which is nothing but sugar in a crystallized form, the solution is allowed to evaporate slowly, and as it cools it forms into those beautiful crystals termed sugar-candy. The threads found in the center of some of the crystals are merely placed for the purpose of hastening the formation of the crystals.

Experiments.

1. Make a strong solution of alum, or of sulphate of copper, or blue vitriol, and place in them rough and irregular pieces of clinker from stoves, or wire-baskets, and set them by in a cool place, where they will be free from dust, and in a few days crystals of the several salts will deposit themselves on the baskets, etc.; they should then be taken out of the solutions, and dried, when they form very pretty ornaments for a room.

2. Fill a Florence flask up to the neck with a strong solution of sulphate of soda, or Glauber's salt, boil it, and tie the mouth over with a piece of moistened bladder while boiling, and set it by in a place where it cannot be disturbed. After twenty-four hours it will probably still remain fluid. Pierce the bladder covering with a penknife, and the entrance of the air will cause the whole mass instantly to crystallize, and the flask will become quite warm from the latent caloric, of which we have spoken before, given out by the salt in passing from the fluid to the solid state. It is better to prepare two

or three flasks at the same time, to provide against accidents, for the least shake will often cause crystallization to take place before the proper time.

Changes of Color Produced by Colorless Liquids.

Make a strong infusion of the leaves of the red cabbage, which will be of a beautiful *blue* color; drop into it a few drops of dilute sulphuric acid, and the color will change to a bright red; add some solution of carbonate of potash, or soda, and the red color will gradually give way to the original blue; continue adding the alkaline solution, and the fluid will assume a bright *green* color. Now resume the acid, and as it is dropped in, the color will again change from green to blue, and from blue to red. Now this simple experiment illustrates three points: first, that acids change the color of most vegetable blues and greens to red; second, that alkalies change most blues and reds to green; and third, that when the acid and alkali are united together, they both lose their property of changing color, and become what is called a *neutral* salt, *i.e.* a compound possessing the properties of *neither* of its constituents.

ACOUSTICS.

Acoustics is the science relating to sound and hearing. Sound is heard when any shock or impulse is given to the air, or to any other body which is in contact directly or indirectly with the ear.

Difference Between Sound and Noise.

Noises are made by the crack of whips, the beating of hammers, the creak of a file or saw, or the hubbub of a multitude. But when a bell is struck, the bow of a violin drawn across the strings, or the wetted finger turned round a musical glass, we have what are properly called sounds.

Sounds, How Propagated.

Sounds are propagated on all bodies much after the manner that waves are in water, with a velocity of 1,142 feet in a second. Sounds in liquids and in solids are more rapid than in air. Two stones rubbed together may be heard in water at half a mile; solid bodies convey sounds to great distances, and pipes may be made to convey the voice over every part of the house.

To Show How Sound Travels Through a Solid.

Take a long piece of wood, such as the handle of a hair broom, and placing a watch at one end, apply your ear to the other, and the tickings will be distinctly heard.

To Show That Sound Depends on Vibration.

Touch a bell when it is sounding, and the noise ceases; the same may be done to a musical string with the same results. Hold a musical pitch-fork to the lips, when it is made to sound, and a quivering motion will be felt from its vibrations. These experiments show that sound is produced by the quick motions and vibrations of different bodies.

Musical Figures Resulting From Sound.

Cover the mouth of a wine-glass, having a foot-stalk, with a thin sheet of membrane, over which scatter a layer of fine sand. The vibrations excited in the air by the sound of a musical instrument, held within a few inches of the membrane, will cause the sand on its surface to form regular lines and figures with astonishing celerity, which vary with the sound produced.

To Make an Æolian Harp.

This instrument consists of a long, narrow box of very thin deal, about six inches deep, with a circle in the middle of the upper side of an inch and a half in diameter, in which are to be drilled small holes. On this side seven, ten or more strings of very fine catgut are stretched over bridges at each end, like the bridges of a fiddle, and screwed up or relaxed with screw-pins. The strings must all be tuned to one and the same note, and the instrument should be placed in a window partly open, in which the width is exactly equal to the length of the harp, with the sash just raised to give the air admission. When the air blows upon these strings with different degrees of force, it will excite different tones of sound. Sometimes the blast brings out all the tones in full concert, and sometimes it sinks them to the softest murmurs.

A colossal imitation of the instrument just described was invented at Milan in 1786 by the Abbate Gattoni. He stretched seven strong iron wires, tuned to the notes of the gamut, from the top of a tower sixty feet high, to the house of a Signor Moscate, who was interested in the success of the experiment; and this apparatus, called the “giant’s harp,” in blowing weather yielded lengthened peals of harmonious music. In a storm this music was heard at a greater distance.

FIREWORKS.

We know full well the intense delight taken by boys in risking their limbs or their lives, especially when such risk is accompanied with noise. Boys always have done so, and always will do so in spite of the very best of advice or precautions. As, therefore, it is impossible to keep them from making noises, and endangering themselves, we have, in this article, endeavored to show them how to make as much noise as possible, with as little danger as possible.

What is there that makes the most noise, and is most dangerous? Gunpowder, of course. Therefore, we have given descriptions of the best methods of employing this material, feeling quite sure that of accidents with gunpowder nine out of every ten are caused by ignorance. We knew a boy who lost the use of a thumb, and took all the skin off the palm of his right hand, by ignorant management of powder. He had read of blasting rocks, and nothing would satisfy him but blasting a bank. So he bored a deep hole in it with a stick, filled the hole with gunpowder, and then poked a lighted lucifer into the powder. The consequence was that his face was so scorched as not to be recognized, all his eyebrows and eyelashes, and most of his hair were burned off, while his right hand was injured, as has been already mentioned. Now that boy had been studiously kept out of the way of powder by female relatives, and was naturally profoundly ignorant of its effects. Had he been taught to handle it, he would not now be forced to keep his right hand closed, or to write by holding the pen between the fingers of his clenched hand.

Gunpowder.

It will not be very advisable for the firework boy to make his own powder, but still it will not be amiss that he should know how it is prepared. Pulverize separately 5 drams of nitrate of potass, 1 dram of sulphur, and 1 dram of newly-burnt charcoal; mix them together in a mortar, with a little water, so as to make the compound into a dough, which roll out into round pieces of the thickness of a pin upon a slab. This must be done by moving a board backwards and forwards until the dough is of a proper size. When three or four of these pieces are ready put them together, and cut them off

into small grains. Place these grains on a sheet of paper, in a warm place, where they will soon dry, but away from a fire. During granulation the dough must be prevented from sticking by using a little of the dry compound powder. This mode of granulation, though tedious, is the only one to be used for so small a quantity for the sake of experiment. In making powder in a large way it is granulated by passing the composition through sieves.

How to Make Touch-Paper.

Dissolve in a little spirits of wine or vinegar a little saltpeter, then take some purple or blue paper, and wet it with the above liquor, and it will be fit for use. When pasting paper on any of the following works take care that the paste does not touch that part which is to burn. The method of using this paper is to cut it into slips long enough to go once round the mouth of a serpent, cracker, etc.

Cases for Squibs, Flower-Pots, Rockets, Roman Candles, Etc.

Procure a hard wooden cylinder, or, if possible, one made of metal, whose diameter corresponds with that of the interior of the proposed case. Roll round it several folds of cartridge paper, and paste the edges well, so that it may be held securely. Tie it round until dry.

To Choke the Cases.

When the cases are thus made they will require to be tied at the lower end. This is called choking them, and as much force is required it is necessary. Fix a wire into a small solid cylinder. Take another short piece, an inch or two long, with a hole up it to admit the other end of the wire, fit it on, and pass it up the case. Then having fastened a piece of whip-cord to a post, wind it round the part left hollow by the wire, which should be about half an inch from the end; pull it tight with the right hand, and work the case round with the left. Cut out a piece of touch paper two inches long, and an inch and a half broad, wind it round the choke, and tie it on with a piece of fine string—twist it to a point. The cases are best choked while damp.

Composition for Squibs, Etc.

Gunpowder, half a pound; charcoal, 1 ounce; brimstone, 1 ounce, or in like proportion; grind them in a muller or pound them in a mortar. Or you may take 1 part steel filings, 1 charcoal, 1 sulphur, and 4 powder, which is a very good mixture, and can be rubbed together in a mortar.

How to Fill the Cases.

Your cases must be very dry when ready, and should be put into an iron or wooden mold; first put in a thimble full of your powder, and ram it down very hard with your ruler, then put in a little more till the case is full, ramming it down hard every time. If you have no mold, hold the case in your left hand with the twisted touch-paper downwards, and fill it after the same manner. When you have filled within an inch of the top, fill up this with loose powder not rammed, for a bang, and fold in the ends; after filling a dozen or two melt some pitch in a small ladle, and smear the end of the case with it by means of a small brush.

To Make Crackers.

Cut some stout cartridge-paper into pieces three inches and a half broad and one foot long, fold down one edge of these pieces lengthwise about three-quarters of an inch broad, then fold the double edge down a quarter of an inch, and turn the single edge back half over the double fold. Open it, and lay all along the channel which is formed by the folding of the paper some meal powder, then fold it over and over till the paper is doubled up, rubbing it down at every turn; this being done bend it backwards and forwards two inches and a half, or thereabouts, at a time, as often as the paper will allow. Hold all these folds flat and close, and with a small pinching cord give one turn round the middle of the cracker and pinch it close; bind it with pack thread as tight as you can, then in the place where it was pinched prime one end and cap it with touch-paper.

When these crackers are fired they will give a loud report at every turn of the paper: if you want a great number of these, you have only to cut the paper longer, or join it on to a greater length; but if they are made very long you must have a piece of wood with a groove in it deep enough to let in half the cracker, which will hold it straight while you are pinching it.

Roman Candles and Stars.

These are best made with the following ingredients: 1 ounce of powder, 1 ounce of sulphur, and 2 ounces of niter. Some persons, however, prefer 1 part sulphur, 1 charcoal, 1 iron filings, 4 of powder, and 8 of niter. The composition being made, in filling the cases fill the contrary way to a squib—stop up the choke by driving down a piece of paper. Put in 1 quill of gunpowder loose and 1 star made in the following manner: 1 ounce of camphor, 1 of sulphur, 2 of meal powder, 1 ounce of the colored fires, moisten them with oil of turpentine, and work them into little round balls. Having placed a star within the case, put in above it 3 quills of the composition, ram down, then powder, star, and composition alternately, till the case is full. Paste touch-paper round the top and twist to a point.

Rockets.

There are several recipes for making rockets, the best of which is 3 ounces of charcoal, 6 of sulphur, 8 of niter, 32 of meal powder. Another very good one is, 3 ounces of iron filings, 4 of powdered charcoal, 8 of sulphur, 16 of niter, and 64 of meal powder. If a smaller quantity is wanted divide each proportion by 2, if a still smaller divide by 4.

Rains.

Sometimes gold or silver rains are added to rockets, which give them a very beautiful appearance. A gold rain is made of 2 parts sawdust, 4 sulphur, 4 meal powder, 6 glass dust, 16 niter, in all 32 parts. A silver rain may be made of 2 parts salt prunella, 8 sulphuret of antimony, 8 sulphur, 8 meal powder, and 14 niter, in all 32 parts.

Catherine Wheels.

These are very pretty fireworks, and are made to turn on a pivot. There are many recipes for the composition of which they are formed; 1 part camphor, 1 sulphur, 1 niter, 2 meal powder. Another is, 3 parts iron filings, 4 sulphur, 12 niter, 16 meal powder. This composition is to be rammed into small cases, and bound round a small wheel having a hole for a pivot in the center.

Various Colored Fires.

The following recipes will give the young firework maker a great variety of the most beautiful fires. They should never be fired in a room, however, and always away from a dwelling.

Crimson Fire.

The principal ingredient in this is nitrate of strontium, of which 40 parts are taken, with 13 of sulphur, 15 of chlorate of potass, 4 of sulphuret of antimony, and 2 of lamp-black. These, as all the ingredients for the other fires, should be rubbed in a ladle, and they may be used in a ladle or iron dish set on the ground.

Blue Fire.

The ingredients of blue fire are 20 parts; 12 of niter, 4 of sulphur, 2 of sulphuret of antimony, and 2 of lamp-black.

Green Fire.

The ingredients for green fire are in 54 parts; 42 of nitrate of barytes, 8 of sulphur, 3 of chlorate of potass, and 1 of lamp-black.

Purple Fire.

The best recipe for purple fire is of 60 parts; 25 of niter, 25 of nitrate of strontium, 7 of sulphur, 2 of realgor, and 1 of lamp-black.

White Fire.

The best and purest white fire is made of 24 parts of niter, 7 of sulphur, 2 of red arsenic, and one of lamp-black.

Spur Fire.

9 parts of niter, 4 of sulphur, and 3 of lamp-black, well rubbed together.

Blue Lights.

These are made of 4 parts of sulphur, 2 of niter, and 1 of powder, and are rammed into squib-cases the contrary way.

Port or Wildfires.

Saltpeter 4 parts, meal powder 6 parts, and sulphur 3 parts. The composition to be moistened with linseed-oil.

Slow Fire for Wheels.

Saltpeter 4 parts, sulphur 2 parts, and meal powder 2 parts.

Dead Fire for Wheels.

Saltpeter 5 parts, sulphur 1 part, lapis calaminaris 1 part, and antimony 1 part.

Cautions.

Such are the principles and methods by which fireworks may be made; but we would advise our young friends to be very cautious, and never to attempt making any fireworks by candlelight; always to select some outhouse for their operations; to see that no iron or steel implements are about the place in which their fireworks are being manufactured, or they may go off before they wish it; to use wooden or brass implements in the bruising, grinding, and sifting of their mixtures; and never to bring the fireworks, or any of their ingredients, into the dwelling-house, or they may suddenly receive a

“Heavy blow and great discouragement.”

To Make an Illuminated Spiral Wheel.

Procure a circular horizontal wheel two feet in diameter with a hole quite through the nave, then take four thin pieces of deal three feet long each, and three-quarters of an inch broad each. One end of each of these pieces is to be nailed to the felloe of the wheel at an equal distance from one another, and the other end nailed to a block with a hole in its bottom, which must be perpendicular with that in the block of the wheel, but not so large. The wheel being thus made, a hoop planed down very thin must be nailed to the

felloe of the wheel, and wound round the four sticks in a spiral line from the wheel to the block at the top; on the top of this block a case of Chinese fire must be fixed, and on the wheel any number of cases, which must incline downwards and burn two at a time. The axis of the wheel must be a little longer than the cone, and made very smooth at the top, on which the upper block is to turn and the whole weight of the wheel to rest.

[THE END.]

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The one footnote has been moved to the end of its section.

Punctuation has been made consistent.

Mathematical notation has been standardized to current conventions. For example, the notation 1-2 for fractions has been changed to $\frac{1}{2}$.

Variations in spelling and hyphenation were retained as they appear in the original publication, except that obvious typographical errors have been corrected.

[p. 14](#): There is no illustration in the original book (This illustration represents)

The following change was made:

[p. 7](#): 2 changed to 3 (No. 3, it)

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